Sample Research Papers

Search Set 1

Key word: IVF outcome prediction machine learning

Paper No: 1

Title:
Comparative study of machine learning approaches integrated with genetic algorithm for IVF success prediction

Year: Published: October 11, 2024

Abstract:
IVF is a widely-used assisted reproductive technology with a consistent success rate of around 30%, and improving this rate is crucial due to emotional, financial, and health-related implications for infertile couples. This study aimed to develop a model for predicting IVF outcome by comparing five machine-learning techniques.

Conclusion:
In this study, we presented a comparative study of machine learning methods integrated with Genetic Algorithm (GA) for predicting the success of In vitro fertilization (IVF) procedures. The study explores the performance of five machine learning algorithms: Artificial Neural Network (ANN), Recursive Partitioning and Regression Trees (RPART), Random Forest (RF), Support Vector Machine (SVM), and AdaBoost, using GA for feature selection.
The standout performer was Adaboost, achieving an accuracy of 89.8% with GA. Random Forest also showed promise, reaching 87.4% accuracy with GA. These results emphasize the significance of feature selection in improving model performance. Overall, the study highlights the effectiveness of ensemble learning methods like AdaBoost and Random Forest in predicting IVF outcomes.

Furthermore, the application of Genetic Algorithm as a feature selection method significantly improves the performance of all classifiers, emphasizing the importance of selecting relevant features for accurate IVF outcome prediction. Ten key features, including female age, AMH, endometrial thickness, sperm count, sperm morphology, follicle size, retrieved oocytes, quality of oocyte and embryo quality, are identified as critical factors influencing IVF success.

These models provide a promising tool for IVF practitioners, allowing for more exact treatment planning. We try to provide a better insight into the IVF success for infertile couples by applying AI and ML, which may potentially decrease emotional and financial difficulties.

Limitations:
This study highlights the ability of the capability of the predicting modelling using the Adaboost as an ensemble approach, combining the predictive performance of the four classifiers (ANN, RPART, RF, and SVM). This approach produces an ensemble model with reasonably high predicted accuracy for successful IVF by combining the mentioned classifiers. Furthermore, the research shows a significant boost in predicting accuracy by combining machine learning techniques with the Genetic Algorithm for feature selection. This novel approach improves the predictability of IVF success, giving significant insights for professionals and patients alike.
As the first limitation for this study, the findings are based on data from a single medical facility in Tehran, Iran. While the size of the data is reasonable, its applicability to different demographics and healthcare settings may be limited.

Finally, the prediction models created in this work should be externally validated in other studies to check the reliability and generalizability.

Method :
The research approached five prominent machine learning algorithms, including Random Forest, Artificial Neural Network (ANN), Support Vector Machine (SVM), Recursive Partitioning and Regression Trees (RPART), and AdaBoost, in

the context of IVF success prediction. The study also incorporated GA as a feature selection method to enhance the predictive models' robustness.

Data set

1.Sample image of dataset.

2, Descriptive statistics of variables for clinical pregnancy and non-clinical pregnancy groups.

Paper No: 2

Title:

Machine learning vs. classic statistics for the prediction of IVF outcomes

Year: Published: 11 August 2020

Abstract:

The study population consisted of 136 women undergoing a fresh IVF cycle from January 2014 to August 2016 at a tertiary, university-affiliated medical center. We tested the ability of two machine learning algorithms, support vector machine (SVM) and artificial neural network (NN), vs. classic statistics (logistic regression) to predict IVF outcomes (number of oocytes retrieved, mature oocytes, top-quality embryos, positive beta-hCG, clinical pregnancies, and live births) based on age and BMI, with or without clinical data.

Conclusion:

Our findings suggest that machine learning algorithms based on age, BMI, and clinical data have an advantage over logistic regression for the prediction of IVF outcomes and therefore can assist fertility specialists' counselling and their patients in adjusting the appropriate treatment strategy.

Paper No: 3

Title: Catalyzing IVF outcome prediction: exploring advanced machine learning paradigms for enhanced success rate prognostication

Year: 05 November 2024

Abstract:
This study addresses the research problem of enhancing *In-Vitro* Fertilization (IVF) success rate prediction by integrating advanced machine learning paradigms with gynecological expertise. The methodology involves the analysis of comprehensive datasets from 2017 to 2018 and 2010–2016. Machine learning models, including Logistic Regression, Gaussian NB, SVM, MLP, KNN, and ensemble models like Random Forest, AdaBoost, Logit Boost, RUS Boost, and RSM, were employed. Key findings reveal the significance of patient demographics, infertility factors, and treatment protocols in IVF success prediction. Notably, ensemble learning methods demonstrated high accuracy, with Logit Boost achieving an accuracy of 96.35%. The implications of this research span clinical decision support, patient counseling, and data preprocessing techniques, highlighting the potential for personalized IVF treatments and continuous monitoring. The study underscores the importance of collaboration between gynecologists and data scientists to optimize IVF outcomes. Prospective studies and external validation are suggested as future directions, promising to further revolutionize fertility treatments and offer hope to couples facing infertility challenges.

Conclusion:

This study represents a significant milestone in the field of assisted reproductive medicine, particularly in the context of predicting *In-Vitro* Fertilization (IVF) success rates. By integrating the expertise of gynecologists with advanced machine learning techniques, we have gained valuable insights that hold great promise for both healthcare practitioners and patients alike. From a gynecological perspective, our findings underscore the critical role of patient demographics and infertility factors in IVF success. Maternal age, years of infertility, gonadotropin dosage, luteinizing hormone levels, and ovarian response have all emerged as key determinants of live birth rates. These insights align with established gynecological knowledge, emphasizing the importance of personalized treatment plans that consider these factors. Moreover, our study highlights the significance of addressing underlying infertility causes when predicting IVF outcomes, such as the positive impact of certain infertility factors like male factor, ovulatory dysfunction, and tubal factor on live birth rates. Furthermore, our exploration of oocyte activation and fertilization rates in relation to IVF success provides valuable clinical insights. Patients with low initial fertilization rates who subsequently received oocyte activation agents showed improved live birth rates, affirming the clinical relevance of such techniques. The identification of specific factors influencing success, including shorter years of infertility, lower gonadotropin dosage, and lower luteinizing hormone levels, empowers gynecologists to make more informed treatment decisions tailored to individual patient profiles. From a machine learning perspective, our study showcases the potential of various algorithms and models in predicting IVF success. Machine learning techniques, including ensemble learning and neural network, have demonstrated accuracy rates ranging from 57 to 96%, depending on the dataset and model choice. Notably, neural network techniques have proven effective in capturing intricate patterns within the data, achieving accuracy rates of approximately 91.44 and 94.71% for the two datasets, respectively. These findings highlight the versatility and potential of machine learning in improving IVF success prediction. Our findings offer gynecologists valuable tools for personalized treatment planning, risk assessment, and treatment optimization. For patients, this means more informed decision-making and better-managed expectations during their IVF journey.

Limitations:

Potential limitations of the study

One potential limitation of the study is its retrospective nature, which hinders the ability to effectively control for unknown confounding factors. Prospective studies, incorporating controlled variables, could offer more robust findings by minimizing the impact of unaccounted variables on the results.

Paper No: 4

Title:

Predictive modeling of pregnancy outcomes utilizing multiple machine learning techniques for in vitro fertilization-embryo transfer

Year:19 March 2025

Abstract:

This study aims to investigate the influencing factors of pregnancy outcomes during in vitro fertilization and embryo transfer (IVF-ET) procedures in clinical practice. Several prediction models were constructed to predict pregnancy outcomes and models with higher accuracy were identified for potential implementation in clinical settings.

Conclusion:
In conclusion, while AI and machine learning hold significant promise for revolutionizing outcomes in IVF-ET, offering powerful tools for both physicians and women with infertility, these technological advances must be carefully managed to address ethical and standardization challenges. Furthermore, this study highlights the need for future research to develop predictive models focusing solely on pre-treatment data to support counseling couples before initiating IVF treatment. Such advancements would enhance the clinical utility of AI models, providing more comprehensive decision-making support and further improving patient outcomes.

Paper No: 5

Title:
Comparison of machine learning classification techniques to predict implantation success in an IVF treatment cycle

Year: November 2022

Abstract:

Research question

Which machine learning model predicts the implantation outcome better in an IVF cycle? What is the importance of each variable in predicting the implantation outcome in an IVF cycle?

Conclusion:
The present study revealed that machine learning algorithms successfully predicted implantation rates in an IVF attempt. In addition, maternal age is by far the most important predictor of IVF success when compared with other variables.

Paper No: 6

Title:
Machine Learning-Based Prediction of IVF Outcomes: The Central Role of Female Preprocedural Factors

Year: **12 November 2025**

Abstract:
Objectives: We aimed to develop and validate a per-cycle prediction model for in vitro fertilization (IVF) success using only preprocedural clinical variables available at the first consultation. **Methods:** We retrospectively analysed 1243 IVF/ICSI cycles (University of Szeged, 21 January 2022–12 December 2023). An Extreme Gradient Boosting (XGBoost version 1.7.7.1) classifier was trained on 14 baseline predictors (e.g., female age, AMH, BMI, FSH, LH, sperm concentration/motility, and infertility duration). A parsimonious 9-variable model was derived by feature importance. Model performance was assessed on the untouched test set and, as a final step, on an independent same-centre external validation cohort (*n* = 92) without re-fitting or recalibration. **Results:** The 9-variable model achieved an AUC of 0.876 on the internal test set, with an accuracy of 81.70% (95% CI 76.30–86.30%), sensitivity of 75.60%, specificity of 84.40%, PPV of 68.60%, and NPV of 88.50%. In external validation, the model maintained strong performance with an accuracy of 78.30%, confirming consistent discrimination on an independent same-centre cohort. Female age was the dominant high-impact feature, while AMH and BMI acted as "workhorse" predictors, and male factors added incremental value. **Conclusions:** IVF outcome can be predicted at the first visit using routinely collected preprocedural data. The model showed consistent discrimination internally and in external validation, supporting its potential utility for early, individualized counselling and treatment planning.

Keywords: **preprocedural factors**; **in vitro fertilization**; **clinical pregnancy**; **live birth**; **machine learning**

Paper No: 7

Title:
Construction and evaluation of machine learning-based prediction model for live birth following fresh embryo transfer in IVF/ICSI patients with polycystic ovary syndrome

Year: Published: 04 April 2025

Abstract:
To investigate the determinants affecting live birth outcomes in fresh embryo transfer among polycystic ovary syndrome (PCOS) patients using various machine learning (ML) algorithms and to construct predictive models, offering novel insights for enhancing live birth rates in this specific group.

Conclusion:
This study developed a live birth prediction model tailored for PCOS fresh embryo transfer cycles, leveraging ML algorithms to compare the efficacy of multiple models. The XGBoost model demonstrated superior predictive capacity, enabling prompt and precise identification of critical risk factors influencing live birth outcomes in PCOS patients. These findings offer actionable insights for clinical intervention, guiding strategies to improve pregnancy outcomes in this population.

Discussion:
In recent years, the correction of metabolic and endocrine abnormalities, combined with low-dose GN ovarian stimulation in IVF, has enabled a growing number of PCOS patients to undergo fresh embryo transfer, thereby reducing the time required to achieve live birth. Utilizing ML methods, this study was among the first to explore the determinants affecting live birth outcomes in PCOS patients after fresh embryo transfer. Seven ML prediction models were developed, each exhibiting strong predictive ability in differentiating live birth outcomes within this group, with the XGBoost model delivering the most favorable performance. This model supports clinicians in initiating early diagnostic interventions for these

patients and offers valuable insights for enhancing pregnancy outcomes in PCOS cases in the future.

ML techniques exhibit enhanced performance over traditional statistical methods when managing complex relationships among numerous features [15–16], as they can identify influencing factors that might be overlooked by conventional approaches based on experience [17]. For predictor selection, LASSO regression and RFE were employed, and their intersection was used to construct predictive models. Among the seven ML models developed, the XGBoost model demonstrated the highest performance, achieving an AUC of 0.822 (95% CI = 0.777–0.867). To further interpret the model and assess the contributions of individual predictors, SHAP analysis was applied to the top-performing XGBoost model. Each SHAP value quantifies the positive or negative impact of a feature on live birth outcomes after fresh embryo transfer in PCOS patients. Among the predictors, blastocyst transfer provided the most substantial contribution to model predictions, showing higher live birth rates in contrast to cleavage-stage embryo transfer [18]. Nevertheless, considering the heightened possibility of ovarian hyperstimulation syndrome in fresh embryo transfers for PCOS patients, some researchers have proposed adopting a freeze-all strategy [19–20]. A randomized controlled trial involving 1,650 patients compared fresh blastocyst transfer cycles with freeze-all and thawed blastocyst transfer groups, focusing on primary outcomes such as singleton live birth rates and secondary outcomes including pregnancy complications, neonatal birth weight, birth defects, and perinatal complications. The findings revealed that the freeze-all strategy significantly enhanced blastocyst implantation rates, live birth rates, and singleton newborn birth weights, contributing to improved maternal-fetal safety and clinical outcomes. However, the study also noted that frozen-thawed single blastocyst transfers were linked to an elevated risk of maternal preeclampsia, raising critical considerations for clinical application [21]. In a recent retrospective cohort study of 10,964 single blastocyst transfer cycles, it was observed that transferring single low-grade blastocysts yielded approximately 30% lower live birth rates relative to 44% for high-quality single blastocysts (with very low-grade blastocysts achieving 14%) without negatively affecting perinatal outcomes [22]. SHAP analysis further demonstrated that transferring two embryos increased the probability of live birth. Although twin embryo transfer yields higher pregnancy rates compared to single embryo transfer [23], existing research and consensus emphasize that it also elevates the risk of multiple pregnancies, along with associated pregnancy complications and adverse perinatal outcomes [24–25]. Consequently, single blastocyst transfer is recommended for PCOS patients, as it balances embryo implantation and live birth success with a reduced risk of multiple pregnancies.

As women age, fertility gradually declines. Once women surpass 35 years of age, the likelihood of spontaneous abortion increases substantially, while pregnancy and live birth rates decrease, accompanied by a heightened risk of various pregnancy and perinatal complications [26]. The connection between maternal age and clinical outcomes following embryo transfer in assisted reproductive technology (ART) has been well-established in numerous studies, with a general consensus highlighting significantly reduced clinical pregnancy and live birth rates in women older than 35 years undergoing ART [27–28]. Consistent findings were observed in this study. Advanced maternal age elevates the risk of early miscarriage, primarily due to a decline in oocyte quality associated with aging, which results in an increased likelihood of embryonic aneuploidy [29]. Preimplantation genetic screening prior to embryo transfer may help mitigate the risk of aneuploidy and reduce the incidence of early miscarriage, although the safety and long-term risks of this approach remain a topic of debate [30]. This study also found that increased progesterone levels on HCG day negatively impacted live birth rates in individuals with PCOS undergoing fresh embryo transfer. Some research suggests that progesterone levels on HCG day may indirectly influence endometrial receptivity and embryo attachment through alterations in gene expression during the implantation window [31]. However, other studies report no association between progesterone levels on HCG day and clinical pregnancy or miscarriage following IVF protocol [32]. As a result, the effect of progesterone measurements on HCG day regarding clinical results post-embryo transfer remains a debated topic.

A high BMI contributes not only to cardiovascular and metabolic disorders but also impairs fertility [33]. Multiple studies have established that a BMI $\geq$ 28 is correlated with reduced ovarian responsiveness and unfavorable pregnancy outcomes [34–35]. In this investigation, BMI $\geq$ 28 was ascertained as a prominent risk factor for the failure to achieve live birth in individuals with PCOS undergoing fresh embryo transfer. The foundational treatment for patients with PCOS involves lifestyle interventions, which encompass a diversified approach including appropriate exercise, dietary control, and behavioral modification. For obese PCOS patients, weight loss has been demonstrated to significantly improve treatment outcomes. Specifically, reducing body weight by 5–10% can lead to notable improvements in ovulation, menstrual cycle regulation, and insulin sensitivity. It is important to emphasize that weight loss should be gradual and sustained over time. HA is recognized as a defining feature of PCOS [36]. However, prior investigations into the influence of HA on reproductive outcomes in PCOS patients have been limited, concentrating mainly on early reproductive stages. Research on subsequent maternal and neonatal outcomes in individuals who achieved clinical pregnancy remains scarce, and the existing literature presents some inconsistencies. In animal experiments, Diao et al. [37] found that high-dose androgen exposure could disrupt endometrial development and interfere with the prostaglandin system, potentially causing early pregnancy loss. Similarly, De Vos et al. [38] reported significantly lower cumulative live birth rates among PCOS patients with HA compared to those without HA. Accordingly, lowering serum testosterone levels in PCOS patients positively influences live birth rates.

Anti-müllerian hormone (AMH) and antral follicle count (AFC) serve as crucial markers for evaluating ovarian reserve function in ART-assisted pregnancy populations. These markers are often utilized as predictors of IVF/ICSI-ET success rates and are regarded as useful metrics for forecasting live birth outcomes. Fertility counseling provided by clinicians frequently relies on changes in these indicators throughout ovarian stimulation to offer personalized guidance [39]. Nevertheless, the findings of this study suggest that AMH and AFC do not exhibit marked predictive value for live birth rates after embryo transfer in PCOS patients, highlighting the necessity for further investigations into the diagnosis and management of PCOS patients undergoing assisted reproductive treatments.

Certain constraints need to be recognized in this investigation. First, the data utilized were obtained from a single center, which may restrict the model’s applicability to patients from other institutions. Second, certain parameters, including insulin and glucose metabolism, were absent from the available electronic medical records. Furthermore, external validation of the constructed prediction model has not been performed, raising concerns regarding its generalizability and highlighting the need for further verification. Moving forward, comprehensive external validation datasets will be gathered to enhance the model's robustness.

Seven ML models were developed in this study to predict live birth following fresh embryo transfer in patients with PCOS, demonstrating strong evaluation accuracy. These models provide critical support for identifying high-risk cases within this population that do not result in live birth, facilitate informed treatment decisions, and enable effective monitoring of patient progression.

Paper No: 8

Title:
Predicting clinical pregnancy using clinical features and machine learning algorithms in in vitro fertilization

Year: Published: June 8, 2022

Abstract:
Assisted reproductive technology has been proposed for women with infertility. Moreover, in vitro fertilization (IVF) cycles are increasing. Factors contributing to successful pregnancy have been widely explored. In this study, we used machine learning algorithms to construct prediction models for clinical pregnancies in IVF.

Conclusion:
Our analysis showed that RF outperformed logistic regression for predicting clinical pregnancy outcomes. In our results, the AUC values of the test dataset with the logistic regression and random forest models were 0.6766 and 0.7208,

respectively. The performance of logistic regression was similar to those of previous studies, and the random forest model outperformed them all.

Discussion:

Significance and contribution of this study

This study attempted to identify variables that could affect clinical pregnancy outcomes and interactions among different variables. Twelve variables were used for analysis in this study. Data were collected from IVF patients between 2007 and 2019 for the model development. Our results demonstrated that 37.88% of women achieved a clinical pregnancy. After developing our predictive model, the random forest model outperformed the logistic regression model for all measures. Among all variables, we found that the "ovarian stimulation protocol" was the most important variable, and long and ultra-long protocols had positive effects on achieving a clinical pregnancy. The second most important variable was the total number of frozen embryos, which had positive effects, peaked at eight, and then began to flatten out.

Data of patients from TMUH were connected to a database of the Health Promotion Administration, Ministry of Health and Welfare, Taiwan. The dataset contained >15,000 IVF cycles with 50 variables. Among these, 42.6% of the "ovarian stimulation protocol" variables in our dataset contained missing values. We compared three functions to impute the missing values. Finally, we incorporated the dataset imputed by the "missForest" function for the analysis.

Another strength of our study is that we provide the importance and propensity of each variable to achieve a clinical pregnancy. Most studies used logistic regression to construct prediction models for pregnancy and live births [19–22], and ORs were used to explain the marginal effects of each variable. Our model comparison results showed that the random forest model outperformed the logistic regression model in terms of accuracy and AUC. In addition, the effects of each continuous variable on the clinical pregnancy probability were nonlinear, and partial dependency plots could be used to illustrate positive or negative effects on clinical pregnancy outcomes. Previous studies have demonstrated that the day of transfer is an important variable associated with pregnancy [19, 23]. In our study, the number of embryos transferred did not distinguish between the day of transfer.

The AUC evaluates the performance of the prediction model. The AUC of the test dataset was 0.7208, and was comparable with other previous studies based on logistic regression with AUCs ranging 0.6–0.7 [7–9]. Our study showed that the random forest model is a good algorithm for classifying clinical pregnancies. Kaufmann et al. reported an accuracy of 58.8% for predicting outcomes of IVF using neural networks [24], and Blank et al. reported that the average AUC, sensitivity, and specificity were 0.74, 0.84, and 0.48, respectively [11].

Clinical interpretations of analytic variables

(1) Female age.

The baseline characteristics of the female age group were significantly correlated with clinical pregnancy outcomes as shown in Table 2 and Fig 2b, and the mean female age in the clinical pregnancy group was younger than that in the group without clinical pregnancy. In the random forest model, the female age was a non-linear curve and showed negative effects on clinical pregnancy outcomes. The probability of a clinical pregnancy dramatically declined after the age of 40 years. Compared to other studies, it is a common predictor in prediction models of clinical pregnancy or live births and has negative effects on clinical pregnancy outcomes [8, 19, 21, 25, 26]. This can be explained by aneuploidy in embryos increasing with maternal age [27], which weakens their developmental potential [28].

(2) Freeze-thawed embryos or fresh embryos.

As shown in Fig 4f, there were more freeze-thawed embryos than fresh embryos. Table 2 shows the rate of pregnancy with freeze-thawed embryos that was 2% higher than that with fresh embryos. In recent studies, the pregnancy rate in frozen embryos was higher than that in fresh embryos [29–32]. Freeze-thawed embryo technology plays a key role in IVF because traditional freezing technology has greatly improved [33]. The number of transfer cycles of freeze-thawed embryos has increased over the past decade, and the technology of freeze-thawed embryo transfer enhanced the live birth rate [34]. Chen et al. found that the advantage of frozen-embryo transfer was correlated with a higher live birth rate and lower ovarian hyperstimulation syndrome risk for infertile women with polycystic ovary syndrome [35]. They also observed that frozen embryo transfer allows the ovary to recover from ovarian stimulation and the estradiol levels. Although there is evidence that freeze-thaw embryo transfer has a significant effect on live births, its effect on clinical pregnancies is uncertain. Wang et al. reported that the rate of clinical pregnancy with freeze-thaw embryo transfer was higher than that with fresh transfer, but the difference was not significant [33]. Our results showed that the freeze-thaw transfer group was more likely to achieve pregnancy than the fresh group. Previous studies reported that the clinical pregnancy rate of the cryopreservation group was significantly higher than that of the fresh group [36, 37]. Shapiro et al. reported that the pregnancy rate with freeze-thawed embryo transfer cycles was greater than that with fresh cycles may be because the typical freeze-thawed embryo transfer cycle uses suboptimal embryos for cryopreservation after their superior siblings are transferred in a fresh state. To keep endometrium steady without ovarian hyperstimulation syndrome (OHSS), intentionally implanting freeze-thaw embryos but not fresh ones to patients in the clinical practice is conducted when the total number of egg retrieval is greater than 20 and estradiol (e2) is greater than 3000. In addition, many freeze-thaw protocols are inadequate to confirm recovered embryo development [37]; therefore, fresh or frozen embryo transfer remains controversial.

(3) Cause of infertility.
Our results showed that the ovary factor, the male factor, and unexplained reasons were more likely to achieve pregnancy through IVF compared to other factors in Fig 4e. In previous studies [21, 38, 39], infertility was found to be the main predictor of clinical pregnancy outcomes. The male factor, ovary factor, and unexplained reasons were obviously more important in clinical pregnancy outcomes than tubal factors, which is in line with a previous study [21]. Infertile women due to male factors can be successfully impregnated with IVF. Other female factors, including recurrent miscarriages and higher prolactin levels, are harmful to clinical pregnancy outcomes. Any of the factors raise the difficulty of becoming pregnant due to interactions. In previous studies, women with unexplained infertility had a higher live birth rate per embryo transfer than those with other causes of infertility [39].

(4) Ovarian stimulation protocols.
Ultralong protocol and long protocol has have been one of the standard protocols for ovarian hyper stimulation. Using Gonadotropin-releasing hormone (GnRH) antagonist for ultralong protocol can tackle cases with endometriosis. GnRH agonist is able to lower then effects of endometriosis, in the meanwhile we are able to perform embryo transfer in the same cycle. In short protocol, the stimulation period is relatively short without pretreatment. Therefore, we believe that the negative effect on short protocol compare to long or ultra-long protocol is without pretreatment with GnRH agonist.

(5) The numbers of embryos transferred.
Higher embryo transfer numbers increase the probability of achieving a clinical pregnancy as shown in Fig 4c. The transfer of more embryos results in more clinical pregnancies. Baker et al. used logistic regression to analyze data and found that a higher number of embryos transferred increased the probability of achieving pregnancy [20].

(6) Duration of infertility.
In our study as shown in Fig 4d, infertile women with more than 1 year of infertility were less likely to become pregnant than those with less than 1 year of infertility, and the duration showed negative effects on clinical pregnancy outcomes. IVF increases the probability of pregnancy and has the best effect on infertility in the first year. A longer duration of infertility is correlated with a reduction in the possibility of conception, but the effects on IVF outcomes remain unclear. Templeton et al. reported a significant reduction in the success rate of IVF due to infertility with a longer duration of infertility [40]. Several studies have reported that the duration of infertility is a predictor of clinical pregnancy. Similar to most studies, the duration of infertility was treated as a continuous variable in our analysis. Ottosen et al. used the duration as a categorical variable to analyze the effects on a pregnancy model. Their results showed that the duration of infertility had a negative nonlinear effect [8]. Compared with the consensus of the duration of infertility in other studies, the probability of clinical pregnancy decreases after the first year [8, 9, 19].

Paper No: 9

Title:
Machine Learning Approach to Predict Clinical Pregnancy Potential in Women Undergoing IVF Program

Year: 2022

Abstract:
Objective: Hidden knowledge could be discovered within a large practical data of in vitro fertilization (IVF) practice. In this study, Machine learning-based data mining techniques were utilized to construct a reliable prediction model for clinical pregnancy in IVF.

Conclusion:
Machine learning algorithm remains effective for the purpose of data mining and knowledge extraction in IVF clinical datasets through which a relatively reliable prediction system for clinical pregnancy could be constructed, provided the available data is sufficient.
The utilization of machine learning algorithms has permitted the extraction of statistical knowledge patterns from a large IVF database. Through the result of our research, both the DT and RF algorithms have a potential in being utilized to build an effective AI-based clinical pregnancy prediction for use in IVF treatment. Each algorithm achieves a similar result, with DT having marginally higher score than RF through the metrics (accuracy, precision, recall, and the F1 score).

Discussion:
This study has displayed the benefits of utilizing machine learning–based data mining concepts to derive knowledge from the considerably large IVF clinical database retrospectively. This study has shown that DT and RF achieved a comparable prediction performance with a balanced accuracy of roughly 0.62 as compared to gradient boost which only achieved a balanced accuracy score of 0.58. DT algorithm is widely used to classify binary outcomes and presents an intuitive set of rules. The optimal combination of variables for each predictor model was

established through GA, and while the sets were different, certain variables were consistently included in all models such as female age, number of top-quality blastocyst(s) and number of top-quality ET. This has corroborated previous studies and served as proof, which implies the significance of female age in predicting pregnancy.

In the present time, classical statistics and machine learning are common complementing tools for the construction of a mathematical model to predict IVF outcomes. However, it has been suggested that machine learning algorithm such as neural network is more powerful in recognizing a broad-range nonlinear association among variables compared to other statistics analysis (Kaufmann et al., 1997). Additionally, a prediction model constructed through machine learning algorithms offers a sufficiently solid forecasting accuracy over the K fold cross-validation method. The potential advantages of machine learning algorithms over classical statistics in attaining true IVF outcome predictions have also been highlighted in a recent study (Barnett-Itzhaki et al., 2020).

Generally, our models displayed sufficient accuracy but not as high as the existing studies (Guh et al., 2011; Hafiz et al., 2017; Hassan et al., 2020). Differing results among studies, including ours, could be attributed by the different data sets and evaluation metrics that were used in each study; thus, comparing the result would be inappropriate. In addition, sample sizes and predictor variables that were introduced to the machine learning algorithms were slightly different. To date, there has yet to exist a consensus protocol that could specify which variables hold the most predictive potential in characterizing a successful IVF cycle (van Loendersloot et al., 2010). This issue is of most importance in machine learning studies because different data sets and diverse variables would produce different sets of rules and prediction models that are unique to each respective data set.

Since the balanced accuracy of the derived models did not achieve at least 0.7, we considered three points of discussion in elucidating these results. First, the noninclusion of important endometrial aspects may carry significant values in distinguishing between pregnant and nonpregnant cases. For instance, the endometrial thickness variable was excluded due to a substantial proportion of unrecorded data before 2017 in our IVF Centre's database. The essential role of endometrial aspects in predicting pregnancy through machine learning algorithms has also been demonstrated (Nanni et al., 2010). In the preliminary study, Nanni et al. attained an area under the curve (AUC) of up to 0.85 by employing sub-endometrial volume, endometrial vascularization index, or flow index combined with female age.

Second, we might not be able to achieve high accuracy when subjects with unspecified infertility were used to predict clinical pregnancy due to the complexity of the multivariable data. In this case, we propose to build a prediction model based on the similarities of the study subjects (e.g., high, normal, and poor responder group) or based on the infertility causes. Another approach is to create the prediction model based on each procedure of the IVF treatment such as an ovarian response prediction model, oocyte retrieval prediction model, blastocyst prediction model, and/or a live birth prediction model. Consequently, noise and vagueness in the relationship between independent and dependent variables might be minimized. Third, the heterogeneity origin of the multicenter-derived data might influence the results of this study.

Certain measures can be taken to solve such issues, for instance by conducting a more selective data selection or introducing new variables that were previously excluded. Referring to existing research to utilize the same set of variables is speculated to offer similar (or better) results, yet the variety of the patient records and the experiment execution to achieve certainty are to be considered. Pregnancy complications are often unexpected and are caused by unexplained factors that were reflected in our dataset. This ultimately challenged the reliability of the prediction model.

Our results possess beneficial inputs for developing countries such as Indonesia. As government subsidies for IVF treatments are nonexistent, creating the prediction model becomes essential for both the patients and clinicians. From an infertile couple's point of view, the availability of such a prediction model could aid in defining a rational expectation of their IVF success rate. Likewise, clinicians could benefit from the calculation to prevent the prescription of unnecessary and overused treatments. Locally, the important findings of this study have presented a novel opportunity to create a supportive decision-making system for our IVF centers.

The strength of this study is reflected in the use of the large, reliable retrospective IVF database to construct the prediction model. The main limitation of this study was the nature of the retrospective data collection. Further research worth investigating would be to evaluate other independent variables that could define the pregnancy outcomes more accurately. Otherwise, developing prediction models appertaining to each step of the IVF treatment may enhance the predictive performance.

Search Set 2

Key word: Artificial intelligence in IVF treatment

Paper No: 1

Title:
Artificial Intelligence in IVF Laboratories: Elevating Outcomes Through Precision and Efficiency

Year: Published: 28 November 2024

Abstract:
Incorporating artificial intelligence (AI) into in vitro fertilization (IVF) laboratories signifies a significant advancement in reproductive medicine. AI technologies, such as neural networks, deep learning, and machine learning, promise to enhance quality control (QC) and quality assurance (QA) through increased accuracy, consistency, and operational efficiency. This comprehensive review examines the effects of AI on IVF laboratories, focusing on its role in automating processes such as embryo and sperm selection, optimizing clinical outcomes, and reducing human error. AI's data analysis and pattern recognition capabilities offer valuable predictive insights, enhancing personalized treatment plans and increasing success rates in fertility treatments. However, integrating AI also brings ethical, regulatory, and societal challenges, including concerns about data security, algorithmic bias, and the human–machine interface in clinical decision-making. Through an in-depth examination of current case studies, advancements, and future directions, this manuscript highlights how AI can revolutionize IVF by standardizing processes, improving patient outcomes, and advancing the precision of reproductive medicine. It underscores the necessity of ongoing research and ethical oversight to ensure fair and transparent applications in this sensitive field, assuring the responsible use of AI in reproductive medicine.

Conclusion:
Integrating artificial intelligence (AI) into in vitro fertilization (IVF) laboratories marks a significant advancement in reproductive technologies. By leveraging AI capabilities, including machine learning, neural networks, and deep learning, IVF

clinics are poised to enhance fertility treatments' precision, efficiency, and outcomes (Table 1).
AI technologies offer transformative potential for IVF laboratories in several key areas:
Enhanced Accuracy and Consistency: AI-driven tools provide more accurate and consistent assessments of embryos, reducing human error and variability in the selection process. This leads to improved success rates in embryo implantation and pregnancy.
Predictive Analytics: AI algorithms excel in analyzing vast datasets to predict treatment outcomes, helping clinicians personalize treatment plans based on predictive insights about patient responses.
Operational Efficiency: Automation and AI-driven processes streamline operations within IVF labs, reducing the manual burden on staff and enabling clinics to handle a higher volume of cases without compromising care quality.
Ethical and Regulatory Adherence: As AI applications in IVF continue to grow, developing ethical guidelines and regulatory compliance remains crucial. Privacy, data security, and informed consent are paramount to maintaining trust and integrity in using AI in reproductive medicine.

7.2. Empirical Support and Research

Numerous studies and real-world applications support the benefits of AI in IVF. For example, research has shown that AI can improve embryo selection accuracy, leading to higher pregnancy rates [82]. Additionally, AI's role in genetic screening can help identify optimal embryos, further enhancing treatment outcomes.

7.3. The Need for Ongoing Research

Despite the promising advancements, continuous research and development are essential to address the challenges associated with AI in IVF, such as algorithmic bias and ethical concerns. Collaborative efforts among technologists, clinicians, ethicists, and regulatory bodies are necessary to develop AI tools that are not only effective but also equitable and ethical.

7.4. Future Directions

Looking forward, the field of IVF will likely witness further integration of AI, particularly in areas like genetic analysis and real-time monitoring of embryo development. These innovations could lead to even more personalized and precise treatments, ultimately improving the success rates of IVF procedures.

In conclusion, AI represents a pivotal development in reproductive technologies, potentially revolutionizing IVF treatments. As this technology continues to evolve, it is imperative to balance innovation with ethical considerations to fully realize AI's benefits in enhancing the quality and success of IVF therapies. This will ensure that advancements in AI not only lead to better outcomes but also adhere to the highest standards of medical ethics and patient care. Such artificial intelligence studies reduce healthcare professionals' workloads, increase their performance, production, and satisfaction in IVF, where teamwork is essential, and contribute to the quality of healthcare institutions by increasing patient satisfaction and healthcare service delivery. Healthcare managers need to be familiar with artificial intelligence studies, which will be used much more in the future, and to include such applications in their institutions. Such artificial intelligence studies reduce healthcare professionals' workloads, increase their performance, production, and satisfaction in IVF, where teamwork is essential, and contribute to the quality of healthcare institutions by increasing patient satisfaction and healthcare service delivery. Healthcare managers need to be familiar with artificial intelligence studies, which will be used much more in the future, and to include such applications in their institutions.

Paper No: 2

Title:
Artificial intelligence in in-vitro fertilization (IVF): A new era of precision and personalization in fertility treatments

Year: March 2025

Keyword:
In-vitro fertilization; Artificial intelligence; Machine learning; Deep learning (DL); Gamete selection; Embryo annotation

Abstract:
In-vitro fertilization (IVF) has been a transformative advancement in assisted reproductive technology. However, success rates remain suboptimal, with only

about one-third of cycles resulting in pregnancy and fewer leading to live births. This narrative review explores the potential of artificial intelligence (AI), machine learning (ML), and deep learning (DL) to enhance various stages of the IVF process. Personalization of ovarian stimulation protocols, gamete selection, and embryo annotation and selection are critical areas where AI may benefit significantly. AI-driven tools can analyze vast datasets to predict optimal stimulation protocols, potentially improving oocyte quality and fertilization rates. In sperm and oocyte quality assessment, AI can offer precise, objective analyses, reducing subjectivity and standardizing evaluations. In embryo selection, AI can analyze time-lapse imaging and morphological data to support the prediction of embryo viability, potentially aiding implantation outcomes. However, the role of AI in improving clinical outcomes remains to be confirmed by large-scale, well-designed clinical trials. Additionally, AI has the potential to enhance quality control and workflow optimization within IVF laboratories by continuously monitoring key performance indicators (KPIs) and facilitating efficient resource utilization. Ethical considerations, including data privacy, algorithmic bias, and fairness, are paramount for the responsible implementation of AI in IVF. Future research should prioritize validating AI tools in diverse clinical settings, ensuring their applicability and reliability. Collaboration among AI experts, clinicians, and embryologists is essential to drive innovation and improve outcomes in assisted reproduction. AI's integration into IVF holds promise for advancing patient care, but its clinical potential requires careful evaluation and ongoing refinement.

Conclusion:

Artificial intelligence (AI), machine learning (ML), and deep learning (DL) have the potential to significantly transform *in-vitro* fertilization (IVF) practices by providing objective, data-driven tools that enhance various stages of the IVF process. These technologies can personalize ovarian stimulation protocols, ensuring that each patient receives the most effective treatment based on their unique characteristics. By optimizing gamete selection through precise assessments of sperm and oocyte quality, AI can improve fertilization rates and embryo viability. Additionally, AI-driven embryo selection can lead to higher implantation success rates, reducing the number of cycles required to achieve pregnancy and thus lowering the emotional and financial burdens on patients [39]. Integrating AI into quality control and workflow optimization further enhances the efficiency and effectiveness of IVF laboratories. AI's ability to continuously monitor and analyze key performance indicators (KPIs) helps maintain high standards and consistent outcomes. Moreover, AI-driven scheduling and resource management

can streamline laboratory operations, minimizing delays and ensuring the timely handling of gametes and embryos.
Despite the promising benefits, the application of AI in IVF must be approached with careful consideration of ethical implications. Ensuring data privacy and security is paramount to protect sensitive patient information. AI algorithms must be trained on diverse datasets to avoid biases and ensure fairness and inclusivity in care. Regular audits and updates of AI models are necessary to maintain their accuracy and mitigate any emerging biases. Continued research and development are crucial to refine AI technologies further and validate their efficacy in clinical settings. Collaborative efforts between AI experts, reproductive endocrinologists, embryologists, and ethicists will be essential to address the challenges and maximize the potential of AI in IVF. By adhering to ethical standards and continuously improving AI applications, the IVF field can offer more effective, equitable, and efficient treatments, ultimately enhancing the overall success rates and patient experiences in assisted reproductive technology.

Paper No: 3

Title:
Artificial intelligence-driven precision treatment of reproductive medicine-related diseases: the optimal protocol choice for IVF-ET

Year: October 2025.

Abstract:
Optimal ovarian stimulation (OS) selection is critical for IVF success, but expert-based decisions often lack consistency in outcomes, cost-efficiency, and personalization, highlighting the need for more individualized and data-driven approaches.

This study propose an artificial intelligence (AI) system that analyzes extensive IVF-ET cycles to uncover OS-pregnancy outcome relationships, enabling personalized treatment recommendations while improving success rates and minimizing unnecessary costs.

This study analyzed anonymized data from 17,791 patients undergoing OS and IVF/ICSI at Tongji Hospital between May 2015 and May 2019. An adaptive AI model was developed to predict key indicators—including progesterone (P), number of oocytes retrieved (NOR), estradiol (E2), and endometrial thickness (EMT) on the hCG day—by integrating personal characteristics, ovarian reserve, and etiological factors. This model

facilitated personalized OS selection, pregnancy outcome grading, and the development of an AI-driven clinical decision support system (CDSS).

An AI-driven clinical decision support system (CDSS) personalizes ovarian stimulation (OS) for IVF-ET. Inputs: baseline demographics, infertility etiology, day-3 labs, and ultrasound. Engine: a data-driven, adaptive ensemble (ACA-FI for feature evaluation, and IRF for prediction) simulates hCG-day P, E2, EMT, and NOR under candidate OS protocols (antagonist, long, ultra-long agonist). Evaluation: predictions are mapped to a pregnancy grading scale (Levels I–IV) and integrated with time–cost metrics. Output: the CDSS recommends the highest-value OS, displaying predicted outcomes in a clinician-friendly dashboard. Impact: enables higher pregnancy probability with lower cost and fewer treatment days.

Discussion:

Despite IVF-ET being a significant breakthrough in treating infertility, pregnancy rate and delivery rate remain low. The low success rate of IVF-ET causes patients' mental and physical damage, economic and time losses. Additionally, the current guidelines do not delineate a recommended regimen for individual IVF-ET cycles. The selection of OS protocols largely relies on the experience of physicians. And discrepancies in OS protocols selection arising from differences in the levels of healthcare facilities and inexperienced practitioners across various reproductive medicine centers may impact the success rates of ART pregnancies. This study established a CDSS that could predict the IVF-ET results and assists the decision of customized optimal OS protocols. The CDSS could be recommended implemented in reproductive medicine centers to help doctors choose the best OS protocols, reduce young doctors' study cycle, increase the cost-effectiveness and pregnancy rates.

The CDSS is established based on large-scale data training and a rigorous scientific prediction process. We established the CDSS based on 13,540 patients at the Reproductive Medicine Centre of Tongji Hospital, and validated and optimized the model in 4,251 patients. First, we identified that E2, P, EMT, and NOR were the top four factors that were in relationship with pregnancy rate among the 27 factors (including personal information, ovarian reserve function and other etiology, OS, follicle and hormone and embryo monitor). The grading criteria for IVF-ET outcomes was built based on the relationship between four indicators and pregnancy rates. Most prior AI applications in ART have focused on laboratory-level prediction tasks, such as embryo viability assessment through time-lapse imaging, sperm selection, or oocyte grading using computer vision and deep learning techniques [53,54]. While these approaches optimize downstream steps of IVF-ET, they do not assist with pre-treatment protocol selection, which has a critical influence on both biological outcomes and patient burden. Our study is among the first to address this gap by introducing a CDSS that evaluates patient characteristics prior to stimulation and recommends personalized OS protocols, thereby shifting AI-driven decision support from the lab to the clinic. This upstream integration complements existing embryo-scoring models and represents a meaningful advancement toward fully individualized ART. Traditional OS selection relies heavily on clinician experience, qualitative interpretation of biomarkers, and institutional norms. While experienced physicians often achieve good outcomes, such approaches are inherently subjective and vary significantly across providers and centers. In contrast, our AI-driven CDSS systematically integrates multi-dimensional clinical data (e.g., AMH, AFC, age, FSH) and applies validated machine learning algorithms to generate consistent, individualized protocol recommendations. Furthermore, the CDSS also demonstrated lower inter-patient variance in cycle cost and gonadotropin dose. Combining with economic cost and time cost, the CDSS provides the maximum support to optimize the outcomes of OS, decreases the cost of trial and error, and improves their patients'adherence and pregnancy results.

The practical application of AI-based CDSS in reproductive medicine requires not only algorithmic accuracy but also seamless integration into real-world clinical workflows. Our CDSS was developed with this clinical operability in mind. The system is designed to interface with existing electronic medical records (EMRs) and outpatient registration systems, enabling automatic data extraction and processing of patients' baseline information (e.g., age, AMH, AFC, hormone profiles) immediately after initial consultation. A user-friendly graphical interface has been developed to provide clinicians with real-time, data-driven OS protocol recommendations. These recommendations are based on the patient's clinical indicators and present a comparative assessment of protocol-specific predicted outcomes—such as E2, P, EMT, and NOR levels on the day of HCG—as well as estimated treatment costs, cycle duration, and efficiency. Clinicians can use this information alongside their own experience to evaluate and select the most appropriate protocol. These recommendations are generated in real time and are intended to serve as clinical references rather than prescriptive instructions. Importantly, the final treatment decision remains under the clinician's authority, allowing for manual override in cases where clinical judgment deems it necessary due to nuances not captured by the algorithm (e.g., rare complications, psychological factors, or patient preferences). In addition, the system also provides predicted pregnancy probabilities, costs, and timelines to help patients form realistic expectations regarding their IVF treatment cycles. Furthermore, we identified key decision points in the workflow—such as protocol selection before stimulation initiation—where the CDSS outputs can be most effectively utilized. The implementation process also considers potential barriers, including clinician resistance to automation, data standardization across EMRs, and the need for physician training. To address these, pilot trials and clinician feedback loops were incorporated during the development phase. Overall, the CDSS is designed not to replace clinical expertise, but to augment it—especially in high-volume centers or settings with less experienced staff—by providing data-driven, evidence-based guidance that enhances decision-making efficiency and standardization.

We innovatively improved an adaptive combination method in the analysis of the relating factors, which can automatically combine different machine learning algorithms, and assign weights by how well fits the data for each method. The combination method fully integrates prediction accuracy of random forest, and the capacity to handle the missing data of XGBoost, while reduces the negative effect of flaw and bias of single method [43]. The integrated analysis of the indicators involved in the whole IVF-ET process of the case cohort provides important information. The ACA-FI ensemble framework is designed to be robust, data-driven, and extensible. Crucially, the framework is adaptive: weights are re-estimated when ported to new cohorts, so the ensemble can rebalance if other learners contribute more in different settings. This approach ensures the framework's adaptability and relevance across different datasets and clinical contexts, reflecting its core purpose as a flexible tool for real-world protocol optimization. Our most research conclusion is consistent with previous conclusion, thereby validating the reliability of the model. Such as, the importance of AMH and age for IVF-ET, the positive correlation of AMH, AFC with embryo quantity and quality, and FSH, age, cycles of IVF-ET are negatively correlating with ovarian reserve indicators and embryo quantity and quality [50,55]. Based on these findings, this study optimizes the OS protocol through AI to improve the outcome of IVF-ET. However, in the current implementation, the methodological framework incorporated only three representative feature selection algorithms—RF, XGB, and SOIL—which may introduce a certain degree of selection bias. Model training and validation were conducted using data from a single large-scale center (n = 17,791), which may raise concerns regarding potential limitations in external validity and generalizability. In future work, we plan to extend the ACA-FI framework by incorporating a broader range of algorithms and validating its performance across multi-center and multi-population datasets to further evaluate its adaptability and robustness.

More interestingly, this CDSS provides us with new significant clinical insights. This study extensively investigates the enduring debate regarding the utilization of GnRH agonists and antagonists protocols. The ESHER recommended antagonist protocol for women with PCOS or predicted normal responders, while both equally for those with poor predicted prognosis [9,56]. According to a *meta*-analysis, the antagonist group had a significantly lower ongoing pregnancy rate than the agonist in IVF-ET cycles. The antagonist has been applied in China since 2014, due to its simplified treatment regimen, reduced medication usage, shortened treatment cycle, and safety and efficacy [[1], [52]]. However, current clinical guidelines provided limited guidance on precisely which patients should receive GnRH antagonist protocols. Against this backdrop, our CDSS predicted individualized pregnancy outcomes under multiple OS options and then recommended the protocol with the most favorable benefit–risk–cost profile for each patient. In our center, the system identified 54.64 % of patients priority to antagonists. Extrapolated to China's exceed one million annual ART cycles [52], this suggests ¥0.54–1.43 billion in yearly savings, easing pressure healthcare services and public health spending. Because pricing varies across institutions and our estimates are derived from a single-center cohort, these figures should be viewed as approximate and interpreted with appropriate caution. These advantages are particularly valuable in low-resource settings and countries where IVF is not fully covered by insurance, thereby enhancing the affordability and accessibility of fertility care. Thus, the CDSS was designed based on the "efficacy-economic-time" comprehensive concept, so that treatment effects and cost are balanced. The "effectiveness – economy – time" CDSS system was applied to the new dataset (n = 4,251). In the test set (n = 4,251), comparing CDSS recommendations with prior physician-selected protocols yielded an ICER of cost saving of ¥2,383 per additional clinical pregnancy, indicating dominance—that is, higher effectiveness at lower cost. Scaled to service volume, the point estimates implyed approximately 60 additional clinical pregnancies and ¥143,000 saved per 1,000 cycles. Future work should extend to cost per live birth and cost-utility analyses.

The study facilitated personalized treatment and precision medicine. There are, however, certain limitations to this study. The primary limitation is the single-center design. Although the cohort is large (17,791 patients over eight years), spaned a broad range of infertility etiologies, and includes all major OS regimens, thereby enhancing representativeness. However, external database verification is still necessary. Thus,we are actively seeking collaborations with additional IVF centers across China and internationally to perform external validation, which will help evaluate the model's robustness and generalizability under different clinical conditions. Methodologically, ACA-FI learns is data-driven and adaptive: when applied to other centers, the weighting and calibration steps are re-estimated on datasets, enabling reducing the risk that single-center idiosyncrasies drive results. To further improve stability and generalizability, we will implement standardized quality controls across participating centers, including ISO 15189–style quality systems for hormone assays [33], ultrasound standardization with operator credentialing and inter-observer calibration, Vienna consensus/Istanbul updates for IVF-lab and embryo assessment [39,40], and ESHRE guidance to harmonize ovarian stimulation protocols [35]. These measures will be embedded prospectively in the planned multicenter validation. Future work will also include a federated learning framework to allow model training across institutions without compromising patient data privacy. Additionally, we acknowledge that not all patient-specific or contextual factors can be fully captured by the algorithm. Clinical decision-making in assisted reproduction is multifaceted and may involve considerations such as prior treatment response, endometrial receptivity history, patient preferences, emotional readiness, and logistical constraints. Our CDSS is therefore not intended to replace physician expertise, but rather to augment pre-treatment planning by presenting data-driven protocol options. It is not designed to suggest or enable in-cycle protocol changes, which remain under exclusive physician control and are rarely feasible in practice. Due to the complex procedure, we also need to optimize the other steps in IVF-ET more than OS. Further, we will focus on more social information and life effects to promote primary prevention of infertility. In conclusion, applications of CDSS will assist clinicians in optimizing OS therapy, predicting pregnancy, and improving IVF-ET outcomes. This will reduce the risk of repeat IVF-ET cycles and associated complications. In summary, the clinical application of this CDSS can help almost all reproductive medicine centers achieve top medical standards effectively, assist young doctors in rapidly gaining experience, and enable patients to receive the most cost-effective ovarian stimulation protocols.

Conclusion:
This AI-assisted CDSS streamlines clinicians' decision-making by enabling efficient and accurate initial judgments on OS, standardizing and personalizing recommendations, and optimizing OS for effectiveness and cost-efficiency.

Paper No: 4

Title:
Omics and Artificial Intelligence to Improve In Vitro Fertilization (IVF) Success: A Proposed Protocol

Year: Published: 21 April 2021

Abstract:
The prediction of in vitro fertilization (IVF) outcome is an imperative achievement in assisted reproduction, substantially aiding infertile couples, health systems and communities. To date, the assessment of infertile couples depends on medical/reproductive history, biochemical indications and investigations of the reproductive tract, along with data obtained from previous IVF cycles, if any. Our

project aims to develop a novel tool, integrating omics and artificial intelligence, to propose optimal treatment options and enhance treatment success rates. For this purpose, we will proceed with the following: (1) recording subfertile couples' lifestyle and demographic parameters and previous IVF cycle characteristics; (2) measurement and evaluation of metabolomics, transcriptomics and biomarkers, and deep machine learning assessment of the oocyte, sperm and embryo; (3) creation of artificial neural network models to increase objectivity and accuracy in comparison to traditional techniques for the improvement of the success rates of IVF cycles following an IVF failure. Therefore, "omics" data are a valuable parameter for embryo selection optimization and promoting personalized IVF treatment. "Omics" combined with predictive models will substantially promote health management individualization; contribute to the successful treatment of infertile couples, particularly those with unexplained infertility or repeated implantation failures; and reduce multiple gestation rates.

Keywords: in vitro fertilization; assisted reproductive techniques; metabolomics; transcriptomics; microRNAs; artificial intelligence; artificial neural network

Conclusion:

To date, various predictive models using demographic and previous cycle parameters, and the evaluation of either metabolomics or transcriptomics, have been evaluated for optimizing IVF outcome, but separately, thus preventing the necessary desired robust results. We propose a system aiming to overcome the lack of accuracy such systems are currently presented with. This is due to the fact that we will combine personal and lifestyle infertile couples' data from different European populations with the most commonly used and accessed "omics" (transcriptomics and metabolomics), interpreted and evaluated through modern mathematical techniques, especially ANNs. After the application of statistical models, the system will be capable of distinguishing, among a large number of demographics, clinical and biochemical parameters, those with the greatest significance for predicting the outcome of an upcoming IVF cycle. Thus, a single device integrating both the molecular tests as well as the outcome of the intelligent systems from the previous cycle could be built. The proposed system is expected to combine high accuracy and efficacy, thus leading to cost effectiveness and benefits to the society and economy. Eventually, we expect the evolution of a simple, reproductive and cheap system that could be used in everyday practice in an IVF unit, especially after the large-scale evaluation of metabolomics and

transcriptomics, which will decrease test costs, thus making them available for routine use. These improvements are to be universally adopted, as this system is expected to receive a wide acceptance as an innovative managerial tool in fertility centers worldwide and could possibly be established in routine clinical practice for the evaluation and management of infertile patients in the near future.
Especially, data will be used both retrospectively and prospectively, in the sense that all new data will serve to continually feed the central database and to synchronously improve the system accuracy. The “Omics” concept will be divided into three phases: 1. analysis of a first sample cohort with different techniques; 2. integration of data and identification of a marker panel for validation; and 3. validation of markers identified during discovery.
We have two final goals: first, the prediction of the IVF outcome at various steps using omics, and second, improved accuracy at the follow-up consultation with the couple after their first IVF failure, providing realistic expectations and informing their next steps to take. The anticipated impact can be divided into two axes: 1. to demonstrate omics’ potential to guide infertility management and 2. to constitute an inexpensive, widely used IVF tool representing an escalated and stratified approach, replacing conventional methods.

Paper No: 5

Title:
Enhancing IVF Outcomes with Artificial Intelligence: Current Advances and Future Possibilities

Year: 26-10-2024

Abstract:
The integration of Artificial Intelligence (AI) into In Vitro Fertilization (IVF) practices has marked a revolutionary shift in reproductive medicine, offering enhanced precision, efficiency, and personalized treatment plans. The rapid advancement of Artificial Intelligence (AI) has led to significant innovations in the field of reproductive medicine, particularly inVitro Fertilization (IVF). Traditional IVF procedures, while effective, often face challenges

such as variable success rates, high costs, and the emotional burden on patients due to multiple treatment cycles. AI offers a promising solution to these issues by enhancing accuracy, personalization, and efficiency throughout the IVF process. AI algorithms have shown remarkable capabilities in diagnosing infertility by analyzing complex datasets from hormone profiles, genetic testing, and medical imaging, enabling early identification of conditions like polycystic ovary syndrome (PCOS) and endometriosis. Moreover, one of the most promising applications of AI in IVF is embryo grading. However, AI systems have been developed to objectively evaluate embryos based on time-lapse imaging, morphology, and other parameters, improving the selection process. Additionally, AI has been instrumental in optimizing ovarian stimulation protocols by analyzing patient data to determine the appropriate medication dosage, minimizing the risk of ovarian hyperstimulation syndrome (OHSS). This review discusses the current state of AI integration in fertility treatments, successful case studies, and ongoing research to develop more sophisticated AI models. Overall, AI holds immense promise in making IVF more accessible, affordable, and successful for patients worldwide, ushering in a new era of precision medicine in reproductive health.Keywords:Artificial Intelligence (AI), In Vitro Fertilization (IVF), Embryo Grading, Predictive Analytics, Robotic Systems.

Conclusion:

The integration of Artificial Intelligence (AI) into In Vitro Fertilization (IVF) has revolutionized the field of reproductive medicine, offering unprecedented opportunities to enhance treatment outcomes, streamline processes, and reduce costs. The capabilities of AI, particularly through machine learning, deep learning, and computer vision, have enabled a more precise, data-driven approach to diagnosing infertility, selecting gametes, grading embryos, and predicting successful pregnancies. By leveraging vast amounts of clinical data, AI systems can identify patterns and correlations that would be difficult for human clinicians to discern, thus facilitating personalized treatment protocols that cater to the specific needs of each patient.One of the key breakthroughs of AI in IVF is its application in embryo selection. Traditional methods rely on subjective assessments by embryologists, which can lead to variability in outcomes. AI-powered systems, however, can objectively analyze embryos based on multiple parameters, including morphology, developmental kinetics, and even subtle features visible through time-lapse imaging. Additionally, AI has proven to be a valuable tool in optimizing ovarian stimulation protocols. By analyzing patient-specific data, including hormone levels, age, and response to previous treatments, AI algorithms can recommend personalized medication dosages, minimizing the risk of complications such as ovarian hyperstimulation syndrome (OHSS). This ensures that patients receive the most effective and safest treatment, reducing both the physical and emotional burden often associated with IVF.In conclusion, AI is poised to redefine the landscape of IVF by making it more accurate, efficient, and patient-centric. The ability of AI to process large datasets, learn from complex patterns, and provide actionable insights has the potential to increase IVF success rates, reduce treatment cycles, and improve patient experiences. While there are challenges that need to be addressed, the benefits of AI-assisted reproductive technologies far outweigh the risks, paving the way for a new era in reproductive health. The continued collaboration between AI developers, clinicians, and regulatory bodies will be essential in ensuring that these technologies are safe, effective, and accessible to all, bringing hope to millions of individuals and couples struggling with infertility

Paper No: 6

Title:
Real-world use of an artificial intelligence–powered clinical decision support tool for ovarian stimulation

Year: September 2025

Abstract:
The AI software used in this study was Stim Assist (Alife Health, Inc., San Francisco, CA). Stim Assist is a clinical decision support software to optimize treatment decisions during ovarian stimulation by providing clinicians with adjunctive predictions about how treatment decisions affect MII outcomes. The

software includes two previously published AI algorithms: a Starting Dose Tool (6) and a Trigger Tool (5). Although the primary focus of this study is on the real-world clinical outcomes and dosing trends observed after the deployment of the AI software, rather than the detailed mechanisms of the Starting Dose Tool and Trigger Tool algorithm development, we have provided a high-level description of each model in the following.

The Starting Dose Tool is intended to be used before the start of stimulation to help guide clinicians on their decision of the starting dose of FSH. The model uses a patient's baseline antimüllerian hormone (AMH) level, baseline antral follicle count (AFC), and body mass index to create a dose-response curve relating the starting dose of FSH to a predicted number of MII oocytes retrieved. The model was trained on a total of 18,591 IVF cycles from three clinics in the United States, between 2014 and 2020. The model uses a K-nearest neighbors (KNN) algorithm to identify the 100 most similar patients to a patient of interest, which are then used to fit a curve relating MII oocytes to the starting dose of FSH. In this model, the total starting FSH dose was calculated as the sum of pure FSH (e.g., Gonal-F or Follistim) plus the FSH component of FSH/luteinizing hormone medication (e.g., Menopur) (6).

The Trigger Tool is intended to be used throughout the stimulation cycle to predict the number of MII oocytes to be retrieved if triggered today, tomorrow, or in 2 days. The Trigger Tool also makes predictions of estradiol (E2) tomorrow, 2 days, and 3 days in the future. The model was trained on 30,278 IVF cycles from three IVF clinics in the United States, between 2014 and 2020. Predictions are made through a set of interpretable linear regression models, which take in a patient's latest E2 and follicle measurements (5).

Stim Assist is intended to be used on patients undergoing conventional IVF cycles for both autologous and donor patients. Stim Assist was fully integrated with the electronic medical record, without the need for manual data entry. When opening the software, a list of patients being actively treated on the current day was automatically populated, with the additional ability to search for specific patients via their patient identification. The AI algorithms generate their predictions in real time (<1 second) to ensure no delays in clinical workflow.

Conclusion:

Our study highlights the abilities of an AI software to help guide clinicians in their decisions regarding FSH dosing and trigger timing throughout IVF treatment. This system was able to help lower starting and total FSH dosing overall than that in patients treated without use of AI guidance. Through this system, clinicians can further individualize treatment for patients and safely optimize ovarian stimulation rather than relying on varied expertise and training alone.

Discussion:

The use of the AI software in this clinical study demonstrated its efficacy in reducing overall starting FSH and total FSH dosing during ovarian stimulation without reducing laboratory outcomes, which was the primary endpoint. Overall, when comparing the treatment arm with the control arm, there was a significant decrease in average starting and total FSH doses, resulting in no significant change in mature egg outcomes. There were no cases of moderate or severe OHSS noted in the study participant's treatment course. The clinicians involved in the study were highly experienced, indicating that the AI software can effectively support even seasoned clinicians in decision-making. Overall, these results suggest that AI could help clinicians optimize their patients' outcomes while reducing medication used, saving money, and helping reduce pharmaceutical waste. This is extremely important in a medical field where access to fertility care is challenging for so many patients, often due to cost.

When separating patients by age groups, there was a significant decrease in average starting and total FSH doses except for the group of patients aged ≥40 years noted in Table 2. Additionally, there were similar average numbers of MII oocytes noted for both the treatment and control groups (Table 1). Although not statistically significant, there was a slight increase in the total average number of MII oocytes in the groups of patients aged 35–40 and ≥40 years (Table 2, Table 4 and 4), along with a slight decrease in patients aged <35 years. These findings suggest that the AI software tool has the potential to individualize FSH dosing, improving the number of MII oocytes collected in age groups with typically lower ovarian response, while promoting safer outcomes in higher responders. However, given the lack of statistical significance, these observations should be further investigated in future studies with larger sample sizes.

It is widely known that baseline AMH level and AFC, along with the patient's age, can be used as a predictive measure of both hyposresponses and excessive responses to ovarian stimulation (7). Additionally, Sadruddin et al. (8) had concluded that AMH is a superior indicator of the ovarian stimulation response, which again was used in our baseline characteristics when selecting patients for comparison in this study. Many clinicians will make treatment decisions with regard to starting FSH dose on the basis of these factors, although it varies by expertise and training. In this study, we used all three of these factors when matching the historical control patients to the treated subjects, and the overall average starting and total FSH doses were significantly lower (Table 1). Therefore, this highlights the further importance of not only individualized FSH dosing but the ability of the algorithm to assist in that process of individualized treatment.

The importance of the starting and overall FSH doses cannot be understated especially when considering ideal ovarian response with stimulation. For those who are considered hyporesponders to ovarian stimulation, higher doses of FSH such as 300–600 IU per day have not shown any real benefit (9). The highest average starting FSH dose in our study was 473.78 IU for patients aged ≥40 years in the treatment arm (Table 2), whereas the overall starting FSH dose for all groups in the treatment arm was 397.09 IU (Table 1), which was lower than that in the control group. Therefore, the results in this study can help pave the way for further use of AI-driven algorithms to help personalize ideal FSH treatment for patients and help identify which patients may benefit from lower doses of medication.

The average number of MII oocytes collected in our study was comparable between the treatment and control groups (Table 1). However, the average numbers of collected MII oocytes were similar at lower FSH doses than in the control. This is further supported by another algorithm model–based study by Correa et al. (10), which showed that most patients with 10–15 MII oocytes collected had FSH dosing between 251 and 300 IU. Although this study did not investigate the postfertilization outcomes such as 2PNs, blastocysts, or live birth rates, as the average number of MII oocytes collected increases, the number of blastocysts and chance of reaching a live birth also increase on average (11). Therefore, the use of MII oocytes as an endpoint is a suitable and meaningful proxy for the effectiveness of ovarian stimulation strategies. Furthermore, postfertilization outcomes will depend on external factors such as sperm quality and laboratory protocol, which are not considered in the AI software, which was designed specifically to optimize the number of MII oocytes. Future studies on this AI software should evaluate the impact on downstream outcomes such as blastocyst formation, euploid rates, and live birth rates to further validate the clinical utility.

Individualized FSH dosing has the ability to not only improve controlled ovarian stimulation outcomes with the number of oocytes collected but also reduce the risks of OHSS. There were no exclusions in this study; thus, there were no separation of those who would be considered poor responders, normal responders, or hyperresponders with assisted reproductive technology. Some data suggest that for those considered hyperresponders (AFC, >15), a 150-IU daily dose of FSH significantly increase the number of oocytes collected and reduce the risk of OHSS, whereas those who are poor or normal responders are recommended to have a starting FSH dose of 150–225 or 225–300 IU (12). The highest starting and total FSH dosing in this study was noted in the group of patients aged ≥40 years, which is to be expected, however, the algorithm still lowered the amount of FSH dosing than that in the control group (Table 2). We also observed no adverse outcomes (moderate or severe OHSS) when using the AI software regardless of the predicted types of responders included throughout the treatment group. Therefore, with the use of the algorithm, patients were not inherently at increased risk of OHSS. Other adverse outcomes, such as lack of blastocyst development or inadequate response to treatment, were not tracked in this study because the primary aim of this analysis was to evaluate the AI software's impact on the number of MII oocytes retrieved and the reduction in gonadotropin usage, rather than broader outcome measures such as blastulation rate or response categorization. These additional adverse outcomes should be the focus of future postmarket analyses.

We analyzed the E2 levels on the day of trigger injection to explore whether differences in dosing were associated with lower E2 levels. No statistically significant differences in the E2 levels were observed between the treatment and control groups. Among patients aged ≥40 years, there was a trend toward lower E2 levels in the treatment group with a higher number of MII oocytes retrieved, although this was not significant. These findings suggest that the AI tool optimizes FSH dosing without substantially altering the E2 levels; however, further studies are needed to determine whether such trends could become significant in larger cohorts. This is particularly relevant to understanding the potential role of

AI-guided dosing in reducing OHSS risk because lower E2 levels have been associated with a reduced likelihood of OHSS (13).
We ran a sensitivity analysis to understand the effects of imputation of missing AMH and AFC in our patient matching analysis by removing all patients from the treatment and control groups with missing baselines. We found that the results were very similar after removing these patients, suggesting that the inclusion of imputed data did not materially impact the overall outcomes and, thus, was a valid approach for patient matching.
The strengths of this clinical study include that the AI software was used in a clinical setting and prospectively in nature, whereas most previous studies evaluating potential AI benefit do so in a retrospective manner without actual tool use. This system also demonstrated its efficacy in reducing overall starting FSH and total FSH dosing during ovarian stimulation while not introducing adverse effects during treatment or reducing clinical outcomes. Additionally, patients in the control group were matched on a 1-to-1 basis to the treatment group using criteria such as, age, baseline AMH, and baseline antral follicles. The limitations of this study include that it was conducted at a single IVF clinic and it was not a randomized, controlled trial. Therefore, future prospective studies using larger populations and a diverse set of clinics should be conducted using this AI software to generate results that can be generalizable to a larger population. Additionally, although the underlying algorithms of the AI software were trained on extensive data sets to incorporate a broad range of patient demographics and clinical practices, it is possible that the training data did not encompass all possible clinical scenarios. It is possible that AI models trained on different data sets could have led to different findings. Lastly, physicians with varying training backgrounds, experience levels, and treatment strategies may interpret the AI-generated predictions differently and use the tool in diverse ways, which should be evaluated in future follow-on studies.

Paper No: 7

Title:
Clinical data-based modeling of IVF live birth outcome and its application

Year: Published: 08 July 2024

Abstract:
Univariate and multivariate analysis shows that 7 factors were closely related to the LBO (with $P < 0.05$): Age, ovarian sensitivity index (OSI), controlled ovarian stimulation (COS) treatment regimen, Gn starting dose, endometrial thickness on human chorionic gonadotrophin (HCG) day, Progesterone (P) value on HCG day, and embryo transfer strategy. By taking the 7 factors as input, the ANN-based and SVM-based LBO models were established, yielding good prediction performance. Compared with the ANN model, the SVM model performs much better and was selected as the final model for the LBO prediction. In real clinical applications, the proposed ANN-based LBO model can predict the LBO with good performance and recommend the embryo transfer strategy of potential good LBO.
The historical data of a total of 1405 patients undergoing IVF cycle were first collected and then analyzed by univariate and multivariate analysis. The statistically significant factors were identified and taken as input to build the artificial neural network (ANN) model and supporting vector machine (SVM) model for predicting the LBO. By comparing the model performance, the one with better results was selected as the final prediction model and applied in real clinical applications.

Conclusion:
We have developed an SVM-based model that can accurately predict the LBO of the IVF, as measured by the indicators of precision, accuracy, sensitivity, and F1 score. This model has been successfully applied in clinical practice to provide precise LBO predictions and as a reliable and scientific tool for guiding decision-making in IVF treatment. For example, it can recommend embryo transfer strategies, optimize COS treatment regimens, and determine the ideal endometrial thickness interval for future embryo transfers. It is of great significance in making treatment decisions, alleviating patients’ psychological burden, and promoting patient compliance throughout the IVF treatment cycle.

Limitations:

This study constructed a novel LBO prediction model based on the retrospective clinical data from 1405 patients. Before constructing the model, univariate and multivariate analysis identified 7 statistically significant influencing factors out of 16: Age, OSI, COS treatment regimen, Gn starting dose, endometrial thickness on HCG day, P value on HCG day and embryo transfer strategy. By taking the 7 screened factors as inputs and the LBO as outputs, two machine learning models, i.e., the ANN-based LBO model and the SVM-based LBO model, were established. By comparing the evaluation indexes of the two models, the SVM-based LBO model demonstrates superior modeling performance, with precision: 77.50% (test set) ~ 86.41% (training set), sensitivity: 75.58%~76.86%, accuracy: 80.43%~84.78%, F1 score: 77.18%~80.63%, AUC: 0.854 ~ 0.912. Therefore, the SVM-based LBO model is selected as the final model for predicting the LBO. Compared with the prevailing research methods, the model proposed in this study demonstrates significantly improved predictive performance.

Our proposed model demonstrates that female age is a significant predictor of LBO. As women age, their ovarian reserve capacity and reactivity decrease, leading to a reduction in the number of oocytes and embryos acquired [2, 3, 6], decreased fertilization ability of eggs and developmental potential of embryos, and an increased proportion of abortion and abnormal birth [7, 8]. In clinical practice, it is advisable for older women to promptly arrange a pregnancy plan and actively engage in pregnancy assistance intervention.

Our finding in Sect. 3 also revealed that the Gn starting dose is an essential factor affecting the LBO. The individualization of Gn starting dose is a standard clinical practice during COS in patients undergoing ART treatment [9], and it is determined based on the patient's age, AMH, AFC, bFSH, BMI, etc. [10]. A small amount of Gn starting dose will lead to insufficient follicle recruitment. However, a large amount will lead to excessive follicle recruitment [11], resulting in an increased incidence of OHSS, and a rise in the progesterone level during COS, ultimately leading to an increase in the cancellation rate of fresh cycle transfer or a decrease in pregnancy rate [12] due to asynchronous endometrial development.

In our proposed model, we have established a robust correlation between the Ovarian Sensitivity Index (OSI), a composite measure of ovarian response [13, 14], and the LBO in patients undergoing IVF treatment for the first time. While previous studies have relied on indicators such as the number of retrieved oocytes, which are often used to reflect ovarian responsiveness, to predict LBO [15], clinical observations have revealed that patients with high ovarian response can still achieve favorable LBO outcomes even with minimal Gn dosage and a smaller quantity of retrieved oocytes [16]. On the contrary, for patients with poor ovarian response (POR), even with increased doses and duration of Gn stimulation and a normal number of oocytes obtained, the live birth rate could still remain low [17, 18]. The concept of OSI serves as a superior measure of ovarian responsiveness to Gn stimulation [14, 19]. In this study, we chose OSI as an indicator due to its comprehensive reflection of both the total dose of Gn and the ovarian response. It allows for simplification in inputting the proposed model without compromising modeling performance.

Results in Sect. 3 also indicate that the COS treatment regimen is a crucial determinant of LBO. In contrast to agonists, antagonists lack the “flare-up” effect and can suppress Gn within a few hours without causing excessive pituitary gland suppression, thereby reducing the required dose and duration of Gn and significantly lowering the incidence of OHSS [20, 21]. Nevertheless, numerous studies have demonstrated that the live birth outcome associated with GnRH antagonist regimens is inferior to that of GnRH agonist long regimens [22,23,24]. In our study, a total of 353 patients were treated with GnRH antagonist regimens, among which only 94 achieved live birth, with a live birth rate of 26.63%, which was much lower than the 55.24% of the GnRH agonist long regimens and the 43.05% of the Ultra-long GnRH agonist regimens. This phenomenon may be attributed to reduced endometrial receptivity in infertile women undergoing GnRH antagonist regimens, leading to a decreased embryo implantation rate [25].

Scholars have suggested establishing a threshold for the P value on HCG day at 1.5-2.0ng/mL [26] due to the potential risk of fresh transplant pregnancy failure associated with higher values [27]. Furthermore, studies have indicated that an increase in P value on HCG day may lead to abnormal expression of endometrial embryo implantation proteins (vascular endothelial growth factor and placental expression factor) and differences in epigenetic profiles, ultimately leading to asynchronous development of the embryo and endometrium and adverse pregnancy outcomes [28]. Elevated P value on HCG day has also been linked to reduced oocyte and embryo quality, leading to lower rates of excellent embryos and cumulative live birth rates [29]. These findings are consistent with the negative partial regression coefficient of P value on HCG day (i.e., B=-0.310) presented in Table 2.

The appropriate endometrial thickness on HCG day is crucial for successful embryo implantation. Some studies have reported that no pregnancy occurs when endometrial thickness is less than 5 mm [30]. Additionally, when the endometrial thickness falls below 7 mm [31, 32] or exceeds 16 mm, it is not conducive to embryo transfer and implantation, which results in a low clinical pregnancy rate [33]. Currently, endometrial thickness of more than 7 mm is considered the conventional lower limit for embryo transfer. Our study suggests that endometrial thickness on HCG day should be regarded as an important index before embryo transfer; endometrial thickness should be adjusted to the ideal thickness before transfer to improve the live birth rate.

Embryos are usually transferred into the womb during the cleavage stage (2–3 days after fertilization) in early times. Prolongation of embryo culture time in vitro to blastocyst stage (5–6 days after fertilization) has been widely employed recently, with the benefits of screening high-quality embryos and keeping the embryo development stage relatively synchronous with the endometrium [34]. Studies have reported that blastocyst transfer yields a higher clinical pregnancy rate and birth rate than cleavage embryos when an equivalent number of embryos are transferred [35,36,37]. The data in Table 1 also demonstrates that the live birth rate following blastocyst transfer is significantly higher than that of embryo transfers at the cleavage stage (1 embryo transfer: 22.42% vs. 15.38%; 2 embryo transfer: 54.34% vs. 46.34%). Two cleaved embryos and one or two blastocyst embryos were independent promotors of clinical live birth compared with one cleaved embryo. It is important to note that the outcome of IVF live birth is not only related to the timing and number of embryos transferred but also the quality of the embryos [38]. However, due to the inherent subjectivity in embryo quality rating and the lack of widespread adoption of time-lapse technology in IVF centers, IVF center rating is not entirely consistent, so we did not include factors of embryo quality rating in this study. This is an aspect that will be explored in future research, utilizing a machine learning algorithm to assess the impact of embryo quality on LBOs. Overall, our findings highlight the significant influence of both stage and number of embryos transferred on the outcome of live birth.

The findings in Sect. 3.4 indicate that the recommended strategy derived from the proposed model may yield superior LBOs for some patients compared to the clinician-based embryo transfer approach. These results have potential implications for informing evidence-based decision-making in IVF clinical practice. Due to the traditional preference for transferring 2 embryos in IVF-ET treatment, the historical data utilized to model the LBO also predominantly supports this strategy, leading to a tendency towards a potential transplant strategy of 2 embryos (the clinical LBO of Strategy-3 is significantly higher than that of Strategy-1 in Sect. 3.4.2). However, scholars believe higher LBOs can be achieved with a single blastocyst transferred (i.e., Strategy-3), and this strategy is becoming the mainstream method of embryo transfer for its decreased multiple pregnancies rate and OHSS risk and increased cumulative pregnancy rate. As the practice of single blastocyst transfer gradually becomes prevalent in current and forthcoming data, the expertise and knowledge encompassed in the newly established model will be continuously updated to align with this mainstream embryo transfer strategy, thereby enhancing the recommended outcomes.

The proposed LBO prediction model in this study is developed by leveraging clinical big data through machine learning techniques. The model considers a comprehensive range of factors, including basic clinical characteristics, COS treatment regimen, OSI, P value on HCG day, endometrial thickness on HCG day, and embryo transfer-related indicators to capture the critical processes in the IVF cycle. Based on the established prediction model, it is possible to forecast LBO by iteratively selecting different embryo transfer strategies and ultimately identifying the optimal strategy based on expected outcomes.

In addition to the recommendation of embryo transfer strategy, the proposed model is also suitable for decision-making on other influencing factors of the LBO model. For example, the model can calculate the endometrial thickness interval, from which the expected live birth situation can be achieved. In this way, whether or not to perform embryo transfer can be determined based on actual endometrial thickness and the prediction LBO. This will be another future work

Paper No: 8

Title:
A new decision-support tool in a multi-center randomized trial for personalized, optimized, and simplified fertility treatment in non-PCOS patients

Year: 16 Sep 2024

Abstract:
This study aimed to evaluate the effectiveness of a clinical decision support tool, Opt-IVF, in achieving the following outcomes: reducing the total cumulative dosage of Gonadotropins (Gns) used during controlled ovarian stimulation cycles and reducing the repeated ultrasonograms (USG) for monitoring follicular growth without compromising the number of good quality blastocysts obtained. The study design employed a multi-center randomized trial. The study enrolled 115 women aged 25–45 years undergoing IVF. Among the participants, 55 were randomly assigned to the intervention group (Opt-IVF), and 60 were randomly assigned to the control group. The intervention involved using a clinical decision support tool, Opt-IVF, to guide Gn dosing and trigger dates. The participants in the intervention group required significantly lower cumulative Gn dosage. The intervention group had higher numbers of oocytes retrieved and M2 retrieved than the control group. The number of good-quality blastocysts, the good-quality blastocyst rate, the ovarian sensitivity index (OSI), and the pregnancy rate in the intervention group were significantly higher than in the control group. The utilization of the clinical decision support tool led to several positive outcomes, including eliminating the need for ultrasound exams after day 5, reducing the dosage of Gn required, and yielding significantly higher numbers of high-quality blastocysts and higher pregnancy rates. Thus, Opt-IVF can successfully provide a personalized, optimized, and simplified approach to superovulation. Opt-IVF consistently outperformed the clinical teams in most of the outcomes.

Conclusion:
The clinical trial demonstrates that with non-PCOS patients, using the Opt-IVF clinical decision support tool for hormone dosing during controlled ovarian stimulation leads to lower overall hormone dosage and eliminates the need for ultrasound testing after D-5 of the cycle. This approach resulted in increased high-quality blastocysts and higher pregnancy rates compared to control patients. Opt-IVF shows promise as a valuable tool for personalized and optimized IVF treatment for patients.

Limitations:
This study presents the outcomes of a multi-center trial of a clinical decision support tool for predicting personalized dosage profiles from D-1 to the end of the controlled ovarian stimulation cycle. The tool is based on a physics-based model (Nisal et al. 2020, 2021) that utilizes follicle size distribution data from D-1 and D-5 to tailor the dosage for each patient. The use of optimal control theory allows for the prediction of an optimal dosage profile, minimizing variance in follicle sizes at the end of the cycle. This study is a multi-center randomized trial with non-PCOS patients to assess the effectiveness of the decision support tool Opt-IVF.

The results obtained in this trial confirm and extend the results recently reported in previous trials (Diwekar et al. 2022, 2023). Importantly, this is a multi-center trial compared to earlier studies from a single center. Opt-IVF-recommended trigger days were used in this trial, unlike previous trials where the clinical investigator determined the trigger day. Unlike in the previous trials, we focused on predicted normal responders. PCOS patients were excluded, and only a small number of poor responders were included. As the intervention group contained a higher proportion of predicted poor responders, we analyzed data for all patients and the normal responders separately. Despite the intervention group having more poor responders than the control group, the intervention group had significantly better outcomes on all metrics except in the average number of embryos, which were similar in the two groups. A possible explanation for this discrepancy is seen in Table 4. In the case of many control patients, embryos were obtained, but in many cases, these embryos did not develop to the stage of good-quality blastocysts, the endpoint in this study. Even among the patients that had at least one good D-5 blastocyst, a higher proportion of the intervention group had multiple good-quality blastocysts, as shown in Table 4. These results suggest that using the decision support tool can increase both the number and the proportion of oocytes capable of maturing after fertilization to a good-quality blastocyst resulting in higher pregnancy rates (Table 4).

Unlike previous studies, where the investigators determined the trigger day based on clinical judgment, we used an Opt-IVF predicted trigger day for all patients in the intervention group in this clinical trial. This may have contributed to the fact that intervention group patients had significantly higher numbers of M2 oocytes than the control group, which was not seen in the earlier trials.

Our results show that using the clinical decision support tool significantly reduced Gn dosage compared to the control. Eliminating the need for ultrasound exams after D-5 of the start of stimulation is also significant. This reduction in testing could lower treatment costs, especially in resource-limited settings, and reduce the logistical and emotional burdens associated with ultrasound testing on patients undergoing controlled ovarian stimulation. None of the participants in the trial experienced OHSS or significant side effects.

One limitation of the study is that we studied mostly normal responders, excluding patients with PCOS and including only a small number of poor responders. In other reported clinical trials (Diwekar et al. 2022, 2023), poor responders and PCOS patients were included in sufficient numbers to show that using the clinical decision support tool resulted in higher numbers of oocytes and high-quality embryos with lower dosages of gonadotropins. Another limitation is that we studied patients from a specific ethnic background (Indian), which might impact the generalizability of the results. However, the follicle development model underlying Opt-IVF was validated in patients from India and the USA (Nisal et al. 2020, 2021). The clinical decision support tool also creates an ovarian follicle development model for a particular controlled ovarian stimulation cycle in an individual patient based on the observed follicular response from D-1 through D-5. This personalized approach essentially eliminates the effects of confounding factors, including age, ethnic background, BMI, and individual variation in ovarian responsiveness to hormones.

Paper No: 9

Title:
Towards Automation in IVF: Pre-Clinical Validation of a Deep Learning-Based Embryo Grading System during PGT-A Cycles

Year :Published: 23 February 2023

Abstract:
Preimplantation genetic testing for aneuploidies (PGT-A) is arguably the most effective embryo selection strategy. Nevertheless, it requires greater workload, costs, and expertise. Therefore, a quest towards user-friendly, non-invasive strategies is ongoing. Although insufficient to replace PGT-A, embryo morphological evaluation is significantly associated with embryonic competence, but scarcely reproducible. Recently, artificial intelligence-powered analyses have been proposed to objectify and automate image evaluations. iDAScore v1.0 is a deep-learning model based on a 3D convolutional neural network trained on time-lapse videos from implanted and non-implanted blastocysts. It is a decision support system for ranking blastocysts without manual input. This retrospective, pre-clinical, external validation included 3604 blastocysts and 808 euploid transfers from 1232 cycles. All blastocysts were retrospectively assessed through the iDAScore v1.0; therefore, it did not influence embryologists' decision-making process. iDAScore v1.0 was significantly associated with embryo morphology and competence, although AUCs for euploidy and live-birth prediction were 0.60 and 0.66, respectively, which is rather comparable to embryologists' performance. Nevertheless, iDAScore v1.0 is objective and reproducible, while embryologists' evaluations are not. In a retrospective simulation, iDAScore v1.0 would have ranked euploid blastocysts as top quality in 63% of cases with one or more euploid and aneuploid blastocysts, and it would have questioned embryologists' ranking in 48% of cases with two or more euploid blastocysts and one or more live birth. Therefore, iDAScore v1.0 may objectify embryologists' evaluations, but randomized controlled trials are required to assess its clinical value.

Conclusion:

Several embryo evaluation tools based on AI technologies have been proposed in IVF to date. For instance, the Spatial–Temporal Ensemble Model (STEM) and its upgrade, STEM+, promisingly reported to be able to predict blastocyst formation with high accuracy and AUC [68]. In a previous work from our group, we reported good consistency between an AI-powered software named CHLOETM and blastocyst quality as defined by clinical embryologists [14]. IVY, a deep learning model producing a score between 0 and 1, was also tested for its prediction of the likelihood of blastocyst implantation and showed encouraging results [50]. Lately, then, some tools have been assessed for a putative prediction of euploidy. Embryo Ranking Intelligent Classification Algorithm (ERICA), for instance, outperformed clinical embryologists in ranking euploid blastocysts as top quality in their cohorts [33], and others such as Euploid Prediction Algorithm (EPA), STORK-A and iDAScore v1.0 itself showed good correlation with euploidy [65,66,69]. Nonetheless, in our view, present and future AI-powered tools should be aimed at supporting embryologists in prioritizing for (either untested or euploid) transfer the embryo(s) more likely to result in a live birth in their cohort, rather than at predicting euploidy (e.g., [70]). The accurate diagnosis of euploidy, as of today, still requires comprehensive chromosome testing technologies—with no report of mosaicism based on ICN—and invasive TE biopsy sampling approaches. Most importantly, although it is essential, euploidy is not sufficient to obtain a healthy baby, and the prediction of this latter outcome—and not of euploidy—should be the main aim of any embryo selection tool. This manuscript summarizes an independent, external, pre-clinical validation of iDAScore v1.0, one of the currently available AI-powered software programs for embryo selection. Within the limitations of retrospective design, our data support iDAScore as a promising tool to objectify embryo evaluation across embryologists and clinics, while preventing time-consuming and potentially biased morphokinetic manual annotations. The current v1.0 model performance upon euploidy and LB after euploid SETs is equivalent to embryologists' performance. Nonetheless, this can be considered a positive result for at least two reasons: (i) iDAScore was not trained to address these outcomes, and (ii) the AUC would be independent from embryologists' subjective assessment and increase objectivity and reproducibility. We also provided preliminary evidence of the current clinical utility of iDAScore v1.0, had it been used for embryo ranking purposes. We advocate for a prospective, possibly multicenter, study to confirm our data with an RCT design. A cost-effectiveness analysis is desirable as well, which should include information about lab workload with and without this tool.

Limitations:

AI and automation will strongly impact the future of IVF, meeting the needs for standardization and lower workload in the laboratories [11,32,51,52]. Nonetheless, AI-powered tools for embryo selection purposes requires further refinement, as most studies in this field show significant and recurrent limitations: (i) the nature of the training datasets is not representative of all clinical practices, (ii) the use of clinical pregnancy or fetal heartbeat as an endpoint rather than LB, (iii) the low sample sizes, and (iv) the lack of multicenter validation data [30,37]. In this study, we aimed at validating iDAScore v1.0 in ICSI cycles with TE biopsy, CCT analysis and vitrified-warmed euploid blastocyst SETs. We assessed: (i) its association with embryo morphology, day of development, euploidy, and LB, and (ii) its putative clinical utility in a retrospective simulation.

As previously reported by other studies, iDAScore v1.0 demonstrated a good correlation with the morphological parameters assigned by experienced embryologists to each blastocyst, either for ICM per se and TE per se, or for overall blastocyst morphology [31,53]. Notably, whenever the ICM and the TE of the same blastocyst were reported of a different quality (e.g., AC versus CA, BC versus CB), iDAScore v1.0 favored the latter. This trend inherently advocates a better predictivity of TE quality upon embryo implantation, as already suggested previously from other groups not using AI-powered tools [54,55,56,57,58,59]. Moreover, embryos achieving full blastocyst expansion on day 5 (<120 hpi) showed higher iDAScore v1.0 than embryos reaching that same stage on day 6 (121–144 hpi) or 7 (>144 hpi), also suggesting a better quality for the former, consistent with our previous analysis based on a different AI tool [14]. These data overall support the use of iDAScore v1.0 to objectify blastocyst morphological assessment within and between IVF clinics, as well as between professionals working at different laboratories, regardless of their experience, social, economic, clinical, and regulatory contexts, potentially influencing clinical choices [16,17,60]. Of note, the performance of iDAScore v1.0 in the evaluation of day 7 and/or blastocysts lower than BB might be suboptimal. In fact, the last frame analyzed by the software in its current version is at 140 hpi, thus it will not capture embryo development on day 7. In addition, poor-quality blastocysts are often deselected by embryologists worldwide, and thus are not sufficiently represented in the training data set. Future versions of iDAScore may benefit from datasets enriched in these populations of embryos and, so far, an early validation of v2.0 already shows significantly improved model performance in comparison to v1.0 with extended image analysis up to 148 hpi [61].

Although a large proportion of excellent quality blastocysts (AA) in a general population of advanced maternal age women are aneuploid (≈50%), while ≈25% of poor quality ones (lower than BB) might be euploid [15], a significant association exists between embryos' morphology and their chromosomal and reproductive competence [6,7,8]. Therefore, we tested iDAScore v1.0 in our dataset for its association with euploidy, to investigate whether this software may play a role in prioritizing euploid blastocysts for ET. Indeed, invasive PGT-A is still the only approach to reliably deselect blastocysts diagnosed with full chromosome aneuploidies and achieve a negative predictive value as high as 98% (i.e., lethality rate when aneuploid ETs were conducted in blinded non-selection studies) [62]. Yet, PGT-A is not universally applicable due to regulations, costs, and expertise; therefore, the long-lasting quest for non-invasive biomarkers of euploidy has also recently focused on AI-powered morphological and morphodynamic assessments [8,63,64,65,66]. Here, we report a significant association between iDAScore v1.0 and euploidy, even after results are clustered according to the day of full blastocyst expansion (day 5–7). The same result is not achieved for excellent, good, average, and poor blastocyst morphology clusters, as defined by conventional embryologists' assessment. This sub-analysis explains why the AUC for euploidy prediction derived from embryologist's evaluation performs better than that of iDAScore v1.0. It must be said, though, that the former approach is limited by its intrinsic subjectivity and limited generalizability [11] as well as by its poor ranking potential (i.e., based on three ICM morphology classes and three TE morphology classes in day 5, 6 or 7 after insemination). Conversely, the latter is more objective and reproducible, and leverages on a score difference as low as 0.1 to discriminate between embryos of slightly different quality. Indeed, among those cycles with at least one euploid and one aneuploid sibling blastocysts (about 50% of the cycles conducted during the study period), iDAScore v1.0 would have ranked euploid embryos as top quality in 63% of the cases versus 47% of the cases for embryologists' assessment. This latter estimate was indeed influenced by 29% of cases where euploid and aneuploid blastocysts were both tied as top quality according to embryologists' assessment.

Of note, iDAScore v1.0 was trained to predict implantation, not euploidy, and although (except for vital aneuploidies) a blastocyst must be euploid to result in a LB, as many as 50% of euploid blastocysts typically fail to implant [20,21,22]. It is therefore remarkable that iDAScore v1.0 showed a more evident association with LB in the context of euploid blastocyst SETs than with euploidy per se, consistent with recent evidence shown by a Japanese group for untested embryos [67]. It is promising that the AUC for the LB outcome mirrored the AUC resulting from embryologists' assessment. It is also reassuring that significantly higher iDAScore v1.0 output was observed for euploid blastocysts resulting in a LB versus reproductively incompetent ones among embryo quality and day of development clusters. These observations may suggest an additional application for iDAScore v1.0, namely it may be the swing vote among sibling euploid blastocysts to rank the embryos for ET. To this end, we conducted a second simulation in all the cycles where at least two euploid blastocysts and at least one LB were obtained (about 20% of the cycles included). This was aimed at understanding how often iDAScore v1.0 would have modified embryologists' choice, had it been used clinically. Notably, a different choice would have occurred in about 50% of the cases, suggesting a concrete influence of this tool, also in the context of PGT-A cycles. According to our data, iDAScore v1.0 would have involved delayed LB outcome in 3% of the cases, and an earlier one in 15% of the cases. However, this simulation may be unbalanced in favor of the software over the operators, because in 59 cases (29%) iDAScore v1.0 putative influence could not be assessed. Specifically, in these cycles the embryologists transferred euploid blastocysts which resulted in a LB, but which were not graded as top quality by iDAScore v1.0, while top scoring blastocysts were not yet transferred at the time of drafting of this paper. On the contrary, the cases where iDAScore v1.0 would have outperformed the embryologists could all be computed, as the incorrect choice of the latter (i.e., no LB achieved) was always evident. Nevertheless, we chose to report these preliminary data here because they represent relevant (although only observational) evidence of the potential contribution of this tool for embryo selection purposes beyond euploidy. An RCT comparing embryologists with iDAScore performance per ET is certainly needed now to assess the true clinical contribution of this tool for embryo selection purposes.

Search Set 4

Key word : Personalized treatment recommendation IVF

Paper No: 1

Title:
Patient-tailored reproductive health care

Year: 18 July 2022

Abstract:
Patient-tailored reproductive health care represents an important challenge for the current practice of infertility prevention, diagnosis and treatment. This approach is based on the concept of precision medicine, taking into account genetic, epigenetic, metabolic and lifestyle characteristics of each individual patient. Even though this goal is still far from being wholly achieved, some aspects can already be put into practice nowadays. Personalization can be based on a comprehensive analysis and synthesis of the patients' personal and familial history, taking into account outcomes of previous assisted reproduction technique (ART) attempts, if available, and confronting these data with the past and the latest clinical and laboratory examination outcomes. As to the male fertility status, there is an urgent need for the inclusion of an accurate diagnostic workup of infertile men leading to the choice of the most adequate follow-up for each particular pathological condition. The follow-up of women who have become pregnant as a result of the ART attempt has also to be personalized. This should be done taking into account both the basic data extracted from the patient's file and those derived from the experience gathered during the latest attempt. Last but not least, the individual condition of each couple has to be taken into account when counseling the patients as to the urgency of the actions to be taken to resolve their fertility problem.

Conclusion:

The current success rates of human assisted reproduction are still rather low as compared with other species. This can be partly due to the inherently lower reproductive performance of the human species as compared to animals. The increasing female age is another factor which comes into play. However, these two aspects should not be blamed as the only culprits. Evidence is accumulating to show that each infertile couple has specific needs which have to be taken into account in choosing the optimal diagnostics, the optimal clinical and laboratory procedures, and the optimal post-treatment follow-up. The more complicated is the case, the more important is the personalized approach. While more high-quality molecular and clinical studies still have to be carried out to introduce precision medicine principles into human assisted reproduction, we already can use information from the patients' history, previous attempts and available genetic and epigenetic data to adapt the treatment to each couple's condition in the optimal way possible.

Limitations:
The limitation of this paper is that the coverage of the issues related to the topic is incomplete because the paper is intended to encourage more authors to publish, including both original research data and more complete reviews. The strength can be resumed as an invitation for using personalized data of each infertile couple to tailor the protocol of their infertility management, instead of having recourse to standard protocols.

Paper No: 2

Title:
Artificial Intelligence in Routine IVF Practice

Year : Published: 26 December 2025

Abstract:
Background: Artificial Intelligence (AI) has emerged as a transformative tool in in vitro fertilization (IVF) as it has done in other sectors. In IVF, AI offers advancements in embryo selection, treatment personalization, and outcome

prediction. It does so by leveraging deep learning and computer vision, as well as AI-driven platforms such as ERICA, iDAScore, and IVY where the goal is to address the limitations of traditional embryo assessment. Key amongst them are the issues of subjectivity, labor intensity, and limited predictive power. Despite rapid technological progress, the integration of AI into routine IVF practice faces key challenges. These are issues related to clinical validation, ethical dilemmas, and workflow adaptation. Rationale/Objectives: This review synthesizes current evidence to evaluate the role of AI in IVF, focusing on six critical dimensions: (1) the evolution of AI from traditional embryology to algorithmic assessment, (2) clinical validation and regulatory considerations, (3) limitations and ethical challenges, (4) pathways for clinical integration, (5) real-world applications and outcomes, and (6) future directions and policy recommendations. The objective is to provide a comprehensive roadmap for the responsible adoption of AI in reproductive medicine. Outcomes: AI demonstrates significant potential to improve the precision and efficiency of IVF. Studies report that AI models can achieve 10 to 25% higher accuracy in predicting embryo viability and implantation potential compared to traditional morphological assessment by embryologists. This enhanced predictive power supports more consistent embryo ranking, facilitates elective single-embryo transfer (eSET) strategies, and is associated with 30 to 50% reductions in embryologist workload per embryo cohort. Early adopters report promising trends. However, large-scale randomized controlled trials have yet to conclusively demonstrate a statistically significant increase in live birth rates per transfer compared to expert embryologist selection. The most immediate and evidenced value of AI lies in hybrid decision-making models. This is where it augments embryologists by providing data-driven, objective support, thereby standardizing workflows and reducing subjectivity. Wider Implications: The sustainable integration of AI into IVF banks on three key aspects: robust evidence generation, interdisciplinary collaboration, and global standardization. To foster these, policymakers ought to establish regulatory frameworks for transparency and bias mitigation. On their part, clinicians need training to interpret AI outputs critically. Ethically, safeguarding patient trust and equity is non-negotiable. Future innovations, mainly AI-enhanced genomics and real-time monitoring, could further personalize care. However, their success depends on addressing current limitations. By balancing innovation with ethical vigilance, AI holds the potential to revolutionize IVF while upholding the highest standards of patient care.

Conclusion:

Artificial Intelligence promises to transform IVF from an artisanal practice into a more precise discipline. This mainly by offering superior capabilities in embryo selection, workflow standardization, and outcome prediction. Existing evidence asserts that AI has the potential to achieve the following: enhance the accuracy of identifying viable embryos by 10–25%, significantly reduce embryologist workload (by 30–50%), and advance elective single-embryo transfer (eSET–defined in Appendix A) through superior, consistent viability prediction. Innovations in AI-enhanced genomics and real-time biosensors could further personalize care. However, the definitive impact on per-transfer live birth rates requires further validation through robust RCTs. This is because the current evidence points more strongly towards gains in process efficiency, consistency, and the reduction in multiple pregnancies.

Despite the promise, AI in IVF necessitates careful adoption. One critical consideration is evidence-based adoption. Retrospective studies such as Rodriguez et al.'s [6] 70,456-embryo analysis demonstrate technical feasibility. However, prospective RCTs proving live birth improvements are urgently needed. It is imperative that AI tools evolve from just predictive curiosities to clinically indispensable assets. This can be achieved through standardized KPIs for cross-platform benchmarking, real-world performance registries and continuous recalibration with local data. The second critical consideration besides evidence-based adoption is ethical vigilance. Key attributes to be fostered include bias mitigation through federated learning across diverse populations to prevent algorithmic discrimination, transparency through XAI frameworks demystifying embryo scoring, and patient sovereignty through clear consent protocols for data usage and AI-assisted selection. The third critical consideration is global collaboration. This needs to be achieved through unified data standards such as HL7 FHIR for IVF to enable interoperable systems, international regulatory harmonization such as FDA and EMA joint SaMD guidelines, and knowledge-sharing platforms such as a WHO-led IVF AI consortium.

The path forward demands augmented intelligence where AI elevates human expertise through hybrid decision-making, continuous upskilling, and patient-centered focus. Simply put, "AI will not replace embryologists, but embryologists using AI will replace those who don't."

As it stands, it is imperative that stakeholders harness AI's power to alleviate infertility suffering. This needs to be accomplished while vigilantly safeguarding human dignity, equity, and the sacred trust inherent in reproductive care. It is only through principled innovation anchored in evidence, ethics, and global solidarity that we can fully ensure these technologies fulfill their transformative potential.

Limitations:

One of the significant challenges undermining the reliability and fairness of AI in IVF is algorithmic bias. This bias often compromises generalizability. One core type of bias underlying the compromised outcomes is data-driven bias, also referred to as demographic bias. Fundamentally, AI models are trained on historical datasets. If these datasets underrepresent certain demographics, the resulting algorithms are likely to perform poorly or produce biased predictions for those groups [38,47,50].

The risk of algorithmic bias is not merely theoretical but has been empirically demonstrated in other high-stakes medical AI systems. A seminal study by Obermeyer et al. [51] audited a widely used commercial algorithm designed to identify patients for extra care. The algorithm was trained to predict future healthcare costs as a proxy for health needs. The audit revealed that at the same algorithm risk score, Black patients had 26.3% more chronic illnesses than White patients (4.8 vs. 3.8 distinct conditions). This significant disparity arose due to systemic inequalities in access. Further, the algorithm systematically underestimated the health needs of Black patients. The authors calculated that correcting this bias would increase the percentage of Black patients identified for extra care from 17.7% to 46.5%. This is a stark illustration of how biased training objectives can perpetuate and amplify healthcare disparities. While comprehensive public audits of IVF-specific AI platforms are still limited, this case highlights the critical importance of scrutinizing the objectives and training data composition of embryo selection algorithms. If such models are trained on datasets that over-represent certain demographics or use proxies for embryo viability that do not generalize equitably, they risk introducing similar biases. This will disadvantage underrepresented patient groups in their pursuit of fertility treatment.

The negative outcomes of this include the exacerbation of existing health disparities and disadvantaging underrepresented populations in access to optimal AI-assisted care [38,50]. The lack of diverse applicability is more evident in models trained primarily on data from high-resource settings or specific genetic backgrounds [11,38]. The second key type of bias is annotation bias. The ground truth, that is, the data used to train embryo selection models is often based on inputs by embryologists, which tend to be subjective. This inherent subjectivity and inter-observer variability [11] means that the training data itself is hallmarked by human bias, which the AI can amplify [11,51]. Further, outcomes are influenced by factors beyond embryo quality such as endometrial receptivity and transfer technique. This creates noisy labels that complicate learning true embryo potential [52,53,54].

The third type of bias is technical and environmental bias. As established earlier, there exist key variations in laboratory protocols, imaging equipment, and image acquisition techniques. These fundamental variances introduce technical heterogeneity [11]. AI models trained on data from one specific technical environment may fail to generalize to others [55]. As such, this limits their broad applicability and requires costly local retraining [11,38,53].

The fourth key issue is opaqueness, otherwise labeled as the black box problem and the lack of explainability. Many complex AI models, particularly deep learning systems, function as black boxes. This makes it difficult to understand why a specific prediction was made [37,38,47,56,57,58]. This lack of explainability hinders trust, making it challenging to identify and correct biases embedded within pertinent models. It also complicates regulatory oversight [37,38,56,58]. The overall effect of this dilemma is that clinicians and patients tend to struggle linking AI decisions with human judgment or biological understanding [46,57].

Overall, it is evident that the lack of explainability fundamentally hinders clinical trust and the ability to identify embedded biases. However, recent advances in Explainable AI (XAI) offer promising pathways to demystify these complex models. Techniques such as SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) can be applied to embryo image analysis. These methods generate visual attribution maps that highlight which specific regions of a static or time-lapse embryo image such as the texture of the cytoplasm or the thickness of the zona pellucida, that most strongly influenced the model’s prediction of viability, ploidy, or implantation potential. By visualizing these ‘attention’ regions, XAI can transform an opaque prediction into an interpretable assessment, allowing embryologists to reconcile AI outputs with established morphological knowledge, identify potential model errors, and build a collaborative, evidence-based decision-making process.

The fifth and last key issue linked with generalizability discrepancies is overfitting and limited external validation. Most models have been shown to have poor external validation [9,11]. This lack of robustness and generalizability is a major limitation for clinical adoption [11,38,53]. Continuous validation across diverse settings is crucial. Nevertheless, it is resource-intensive [38,53]

Search Set 5

Key word: Explainable AI in IVF prediction

Paper No: 1

Title:
Explainable artificial intelligence to identify follicles that optimize clinical outcomes during assisted conception

Year: Published: 08 January 2025

Abstract:
Infertility affects one-in-six couples, often necessitating in vitro fertilization treatment (IVF). IVF generates complex data, which can challenge the utilization of the full richness of data during decision-making, leading to reliance on simple 'rules-of-thumb'. Machine learning techniques are well-suited to analyzing complex data to provide data-driven recommendations to improve decision-making. In this multi-center study (n = 19,082 treatment-naive female patients), including 11 European IVF centers, we harnessed explainable artificial intelligence to identify follicle sizes that contribute most to relevant downstream clinical outcomes. We found that intermediately-sized follicles were most important to the number of mature oocytes subsequently retrieved. Maximizing this proportion of follicles by the end of ovarian stimulation was associated with improved live birth rates. Our data suggests that larger mean follicle sizes, especially those >18 mm, were associated with premature progesterone elevation by the end of ovarian stimulation and a negative impact on live birth rates with fresh embryo transfer. These data highlight the potential of computer technologies to aid in the personalization of IVF to optimize clinical outcomes pending future prospective validation.

Discussion:
We present findings from a large multi-center European study incorporating data from 19,082 patients utilizing XAI techniques to establish the relationship between follicle

sizes at the end of ovarian stimulation (OS) and the subsequent retrieval of mature oocytes, as well as relevant downstream clinical outcomes. We showed that a contiguous range of follicles sized 13–18 mm contributed most to the number of mature oocytes subsequently retrieved. In comparison to using lead follicle size to inform trigger timing, our data suggests that maximizing the proportion of follicles within this size range could further optimize the number of mature oocytes retrieved to improve IVF outcomes, including LBR, pending prospective evaluation. Furthermore, extending the duration of OS resulted in a greater number of larger follicles (>18 mm) on the DoT that not only contributed less to the yield of mature oocytes but also resulted in premature progesterone elevation with a consequent negative impact on live birth[17,18], likely due to its adverse effect on the endometrial stage[14]. These data highlight the potential of ML methods to aid in personalizing ART treatment to optimize clinical outcomes.
Our results are consistent with a previous pilot study using a much smaller sample size of 499 patients that found that follicles sized 12–19 mm on DoT were most contributory to the number of mature oocytes[10]. Hariton et al. found that an input feature containing the number of follicles sized 16–20 mm on the DoT was most contributory to the model performance using an ensemble ML model of similar complexity in data from a single clinic comprising 7866 ICSI treatment cycles with various protocol types[19]. Similarly, Reuvenny et al. used an XGBoost model on data from GnRH antagonist (“short” protocol) co-treated cycles from a single center (n = 3599 treatment cycles) to show that an input variable of 14–16 mm sized follicles on the DoT highly contributed towards model performance[20]. Further, Fanton et al. developed a linear regression model with data from three clinics in the USA (n = 30,278 treatment cycles) that identified an input variable of 14–15 mm sized follicles on the DoT as the most important, followed by 16–17 mm follicle sizes[21]. In our study, we examined individual follicle sizes to identify the group of follicles that contribute most to the retrieval of mature oocytes and, in turn, downstream clinical outcomes.

To validate the methodology further, we also analyzed follicle sizes that were most important on the days prior to DoT. We showed that follicles sized 10–15 mm on DoT-2 and 12–17 mm on DoT-1 were the most contributory to predicting MII oocyte yield on the days preceding DoT. This smooth trajectory of optimal follicle size range shift corroborated well with expectations in mean follicular growth rates per day during OS[16]. The aforementioned ML studies also showed agreement, where the most important input variables to the models utilizing follicle sizes dropped by 1–2 mm in range a day prior to the DoT[20,21].

In order to examine whether similar follicle sizes were important in predicted good ovarian response patients, we stratified our data by age. We found that follicles sized 13–18 mm were most contributory in younger patients aged ≤35 years, but those 11–20 mm (particularly 15–18 mm) were most important in patients >35 years. In line with our data whereby increased mean follicle size by the end of OS was associated with a negative impact on LBR, it has been suggested that older patients with diminished ovarian reserve may benefit from earlier trigger administration in modified natural cycles[22]. To date, the impact of age on trigger timing in IVF cycles remains uncertain. Likewise, although most patients are required to be non-smokers to access state-funded care, there was insufficient recording of smoking status to formally assess any impact of smoking.

It has been hypothesized that due to the preceding period of gonadotropin suppression that could synchronize follicle growth, the GnRH agonist co-treated (“short”) protocol could lead to a more uniform cohort of follicles than in the GnRH antagonist co-treated (“long”) protocol[23]. We found that larger follicles sized 14–20 mm were more contributory to the yield of mature oocytes in GnRH agonist co-treated protocol cycles whereas follicles sized 12–19 mm were most important in GnRH antagonist protocol cycles. A meta-analysis of seven randomized control trials ($n = 1295$ patients) comparing protocol types found that patients undergoing the “long” protocol had a significantly higher number of oocytes retrieved ($p < 0.00001$) in those that received a delayed hCG trigger by 24–48 h after OS[24]. Our data are consistent with these previous trials, suggesting that larger follicles may be associated with improved oocyte yield in the GnRH agonist co-treated protocol[25,26].

Several points of strength should be highlighted in our presented study. Firstly, we used XAI techniques in our study. Explainability is currently a paramount characteristic of AI decision-making in ART clinics and the introduction of data-driven insights that align with clinical reasoning promotes the possible wider adoption of clinical decision-support systems in the ART domain[2,27,28]. Secondly, the use of data across two countries comprising eleven clinics presents a heterogeneous patient population with a variety of clinical practices and treatment protocols. Since many clinics are involved, the use of internal-external validation at this stage of development was appropriate[29,30], whereby the developed models and their performance metrics are provided with a standard deviation due to each clinic behaving as an independent test set to observe permutation importance and model error. Furthermore, we chose to avoid random splits of data from the same clinics in both training and test sets when reporting outcome metrics, as this has been shown to result in bias and unknown generalizability ("transportability") to wider patient populations[30]. Thirdly, by only using individual follicle sizes as separate input variables (i.e., not in a grouped/binned fashion), a contiguous range of follicles that contribute most towards mature oocyte yield was more explicitly identified. Finally, we ensured only to include the first treatment cycle of each patient in this study for model development and validation (i.e., $n = 19{,}082$ represents both the number of treatment-naive patients and cycles). This was to avoid auto-correlation between successive cycles of a single patient (e.g., a clinician's decision-making is likely to be influenced by a previous treatment cycle) since longitudinal random effects across sequential cycles can influence the permutation importance[31]. Similarly, input parameters that present multi-collinearity (e.g., estradiol and follicle size[32]) can also result in unreliable insights from permutation importance analysis.

Variability in follicle size measurements, both within and between observers, has been a documented challenge in ART[33]. We observed a few cases where the total number of oocytes retrieved exceeded the follicles recorded at the time of trigger. This discrepancy underscores the likelihood that smaller follicles are often not consistently recorded, a practice echoed in anecdotal discussions with clinicians, who frequently attribute this to the lower probability of such follicles yielding oocytes. A previous study has shown that follicle imputation has limited impact on enhancing model performance in this specific context[21]. Although excluding such treatment cycles could artificially improve model error, it is then not possible to build models that are more robust to such measurement error in future applications. Our approach utilizes an ensemble-based ML model with Bayesian optimization and can mitigate against the impact of data inconsistencies due to extremes of biological variation and/or measurement error[33]. By employing a loss metric like mean absolute error (MAE), which is less sensitive to outliers, the model offers a more robust analysis validated across multiple clinics. The need for more objective ultrasound scanning methods, potentially through the integration of automated algorithms, may further improve the accuracy and reliability of follicle measurements and associated algorithms in ART[33].

Our data suggest that a novel approach to deciding when to administer the trigger of oocyte maturation could be based on the proportion of intermediately-sized follicles (e.g., 13–18 mm) rather than the traditional threshold-based approach assessing when 2–3 lead follicles reach 17 or 18 mm in size (Fig. 4). Further, our data also suggests that mean follicle size could impact LBR in fresh embryo transfer cycles (Fig. 5), aside from a direct effect on the ability to yield oocytes, potentially via larger follicles resulting in premature progesterone elevation[34,35]. A prospective randomized controlled trial is required to demonstrate the benefit of our new proposed approach to determine trigger administration based on the entire follicle cohort in comparison to the current threshold-based approach. Since our data demonstrates a possible trade-off between LBR and delayed trigger administration, a trial comparing trigger strategies on this basis would therefore be potentially more definitive in determining the impact of trigger timing on clinical outcomes. It should be noted that although a range of follicle sizes that contribute relatively more than others varied marginally depending on the patient stratifications considered (Fig. 2). Ultimately, an ML model that considers individual follicle sizes and their relative contributions, in addition to patient characteristics, could be harnessed as part of a clinical decision support system[2,28].

In conclusion, we establish that intermediately-sized follicles on the day of trigger contribute the most to the retrieval of mature oocytes and subsequent embryo development. Utilizing the sizes of all follicles, rather than just the size of only the lead follicles, could offer a target for OS protocols and inform the timing of trigger administration to optimize clinical outcomes. These data highlight the potential of XAI techniques to provide data-driven optimization of IVF treatment to improve clinical outcomes

Paper No: 2

Title:
An explainable ultrasound-based machine learning model for predicting reproductive outcomes after frozen embryo transfer

Year: 8 May 2025

Abstract:
Research question
Can an optimal machine learning model be developed to predict reproductive outcomes following frozen embryo transfer (FET)?

Design:
This prospective study included 787 infertile females who underwent FET. The participants were split into a training cohort (*n* = 550) and a test cohort (*n* = 237) at a ratio of seven to three. Radiomics features were extracted from ultrasound images of the endometrium and junctional zone. A radiomics model was developed to generate the radiomics score (rad score). Logistic regression was applied to process the clinical data and create a clinical model. A fusion machine learning model was developed by integrating the rad score with independent clinical data using the XGboost algorithm. The performance of the models was compared using the area under the receiver operating characteristic curve (AUC). The SHapley Additive exPlanations (SHAP) method was used to interpret and visualize the contributions of features to the outcomes of FET.

Result:
The fusion model demonstrated superior performance, as indicated by an AUC of 0.861 (95% CI 0.829–0.890), in the training cohort, surpassing both the clinical model (AUC 0.680, 95% CI 0.635–0.722; $P < 0.001$) and the radiomics model (AUC 0.814, 95% CI 0.777–0.848; $P < 0.001$). The SHAP summary plot reveals the impacts of each feature on the predictive model, and the rad score was found to be the main feature. SHAP force plots provided explanations at the individual level.

Conclusion:

A new clinical ultrasound-based radiomics model was developed that provides a new direction for predicting the occurrence of clinical pregnancy following FET. By explaining the model via the SHAP method, clinicians may better comprehend the contributors to the outcomes of FET in individual patients, and make better decisions before FET. More multicentre prospective studies are expected to obtain more robust evidence for the clinical use of this ultrasound-based radiomics model in subsequent studies.

Discussion:

This study represents the first prospective analysis to integrate radiomics features of endometrium and junctional zone with clinical data to predict the pregnancy rate following FET via machine learning models. This study initially constructed three models: a radiomics model incorporating ultrasonographic features of endometrium and junctional zone; clinical model A utilizing clinical data; and fusion model A integrating all information. Subsequently, after the exclusion of embryo quality as a factor, two additional models were built without embryo classification: clinical model B and fusion model B. Fusion model A demonstrated superior performance in both the training and test cohorts compared with the other models. Additionally, the study elucidated fusion model A, and visualized the significance of the models via the SHAP method, with the rad score identified as the top feature.

In IVF-embryo transfer cycles, a significant proportion of adverse outcomes can be attributed to suboptimal endometrial receptivity (Achache and Revel, 2006). Researchers have long sought to identify ultrasonic parameters that could serve as predictors for endometrial receptivity. While some studies have suggested that factors such as EMT and pattern, EMV and blood flow parameters may impact endometrial receptivity (Elsokkary et al. 2019; Mercé et al., 2008; Yuan et al., 2016), one systematic review characterized the ultrasonic assessment of endometrial receptivity as having limited predictive ability (Chen et al. 2021). Liao et al. (2022) utilized 3D ultrasound to assess endometrial receptivity, and reported that EMT and VFI were independent factors related to clinical pregnancy; although VFI was not significant in the present study, the results regarding EMT were consistent with the findings of Liao et al. The discrepancies in results may be attributed to observer variability, variations in the timing of ultrasound scans, differences in sample size, and diverse transfer protocols. Nevertheless, ultrasound-based parameters may not be adequate for assessing the complexity of endometrial receptivity.

Radiomics, which uses high-throughput algorithms to extract quantitative features from images that are imperceptible to the human eye, may offer a more comprehensive assessment and develop a model via machine learning methods to analyse and predict the image features in more depth (Zheng et al., 2020, 2023). In a previous study by Liang et al. (2023), women who underwent FET with high-quality embryos were selected as research subjects. The study used data on sex hormone concentrations during menstruation and ultrasonic measurements on the progesterone initiation day as clinical data; it combined these data with TVUS data on EMT on the progesterone initiation day to construct a multimodal fusion model, and AUC was 0.825. However, the authors did not set up training and test groups, and the type of blastocyst was not considered. It is widely recognized that the type of blastocyst and patient age can impact the outcomes of embryo transplantation significantly (Pathare et al., 2023; Teh et al., 2016b). Moreover, researchers have favoured use of the junctional zone in recent years. Some researchers believe that the junctional zone changes periodically with hormone concentrations and is associated with endometrial peristalsis, and that, therefore, it is closely related to the success of FET (Tanos et al., 2020). In the study by Liu et al. (2021), the junctional zone volume/EMV ratio was analysed to evaluate clinical pregnancy, and the results suggested that a smaller junctional zone volume/EMV ratio was associated with a higher clinical pregnancy rate. Hence, in the present study, ultrasound-based data, age and embryo grade were utilized to establish a clinical model. The endometrium and junctional zone on ultrasonography were employed as ROI to develop the radiomics model. The optimal fusion model achieved an AUC of 0.861 in the training cohort and 0.793 in the test cohort. Otherwise, a comparative analysis was conducted between fusion models with and without embryo quality information. The results indicated that the models incorporating embryo quality information exhibited superior performance compared with those that did not include such information. This underscores the significance of embryo quality in predicting the outcomes of FET. In addition, embryo quality was used as an independent variable to conduct a stratified analysis of the optimal model. The results showed that the fusion model performed well across different grades of thawed embryos, demonstrating the stability of the model. To explain the fusion model, the main predictors of the outcomes of FET were determined using a SHAP summary plot and, for individual females, via SHAP force plots. The results suggested that the rad score, embryo grade, age and EMT were major features for the fusion machine learning model to predict the outcomes of FET.

However, this study has various limitations that need to be addressed. First, the machine learning model in this study was trained and tested in a single reproductive centre, and samples from multiple centres are needed for further validation. Second, the sample size of individuals undergoing non-high-quality embryo transfers was insufficient. In a future study, the authors will incorporate a larger number of eligible participants from this category. Third, to control a possible variable, the participants in this study received FET, not fresh embryo transfers, although fresh embryo transfers are worth studying. In addition, the ROI was drawn manually on ultrasound images, which is time-consuming and may introduce subjective bias. With the advancement of machine learning algorithms, the use of automatic segmentation methods and classification tasks can be investigated further. Finally, clinical pregnancy was based on visible fetal cardiac tube beats in the first trimester, and the final live birth rate needs further study.

Paper No: 3

Title:
Interpretable, not black-box, artificial intelligence should be used for embryo selection

Year: 02 November 2021

Abstract:
Artificial intelligence (AI) techniques are starting to be used in IVF, in particular for selecting which embryos to transfer to the woman. AI has the potential to process complex data sets, to be better at identifying subtle but important patterns,

and to be more objective than humans when evaluating embryos. However, a current review of the literature shows much work is still needed before AI can be ethically implemented for this purpose. No randomized controlled trials (RCTs) have been published, and the efficacy studies which exist demonstrate that algorithms can broadly differentiate well between 'good-' and 'poor-' quality embryos but not necessarily between embryos of similar quality, which is the actual clinical need. Almost universally, the AI models were opaque ('black-box') in that at least some part of the process was uninterpretable. This gives rise to a number of epistemic and ethical concerns, including problems with trust, the possibility of using algorithms that generalize poorly to different populations, adverse economic implications for IVF clinics, potential misrepresentation of patient values, broader societal implications, a responsibility gap in the case of poor selection choices and introduction of a more paternalistic decision-making process. Use of interpretable models, which are constrained so that a human can easily understand and explain them, could overcome these concerns. The contribution of AI to IVF is potentially significant, but we recommend that AI models used in this field should be interpretable, and rigorously evaluated with RCTs before implementation. We also recommend long-term follow-up of children born after AI for embryo selection, regulatory oversight for implementation, and public availability of data and code to enable research teams to independently reproduce and validate existing models.

Conclusion:
We do not aim to demonize AI. Quite the opposite, we enthusiastically acknowledge that it has the potential to radically enhance IVF. AI in IVF has the potential to help couples have children earlier in their treatment and at a lower cost. The point is that AI algorithms are portrayed as applications which are so clever that they can detect order from background noise that is too subtle for the human to detect. On the flip side is the risk that the algorithm will find order when none exists or is there by association. These are two sides of the same coin. Researchers, companies and clinics must ensure that the technology they promote or adopt brings real, measurable benefits to patients and, most importantly, does not expose

them to unreasonable risks. We have highlighted limitations of current ML models, we drew specific attention to the ethical concerns associated with AI in IVF, and suggested changes to design and evaluation. We have argued that there should be interpretable ML models that clinicians can understand, troubleshoot and explain to their patients, rigorously evaluated with RCTs. Black-box AI risks a new machine paternalism.

Paper No: 4

Title:
Beyond black-box models: explainable AI for embryo ploidy prediction and patient-centric consultation

Year: Published: 04 July 2024

Abstract:
To determine if an explainable artificial intelligence (XAI) model enhances the accuracy and transparency of predicting embryo ploidy status based on embryonic characteristics and clinical data.
This retrospective study utilized a dataset of 1908 blastocyst embryos. The dataset includes ploidy status, morphokinetic features, morphology grades, and 11 clinical variables. Six machine learning (ML) models including Random Forest (RF), Linear Discriminant Analysis (LDA), Logistic Regression (LR), Support Vector Machine (SVM), AdaBoost (ADA), and Light Gradient-Boosting Machine (LGBM) were trained to predict ploidy status probabilities across three distinct datasets: high-grade embryos (HGE, n = 1107), low-grade embryos (LGE, n = 364), and all-grade embryos (AGE, n = 1471). The model's performance was interpreted using XAI, including SHapley Additive exPlanations (SHAP) and Local Interpretable Model-agnostic Explanations (LIME) techniques.
The mean maternal age was 38.5 ± 3.85 years. The Random Forest (RF) model exhibited superior performance compared to the other five ML models, achieving an accuracy of 0.749 and an AUC of 0.808 for AGE. In the external test set, the RF model achieved an accuracy of 0.714 and an AUC of 0.750 (95% CI, 0.702–0.796). SHAP's feature impact analysis highlighted that maternal age,

paternal age, time to blastocyst (tB), and day 5 morphology grade significantly impacted the predictive model. In addition, LIME offered specific case-ploidy prediction probabilities, revealing the model’s assigned values for each variable within a finite range.

Conclusion:

The model highlights the potential of using XAI algorithms to enhance ploidy prediction, optimize embryo selection as patient-centric consultation, and provides reliability and transparent insights into the decision-making process.

Search Set 6

Keyword: Machine learning infertility treatment outcome

Paper No: 1

Title:
Artificial Intelligence, Deep Learning, and Computer Vision in Hysteroscopy: A Systematic Review

Year: Published: 18 June 2026

Abstract:
Background/Objectives: Hysteroscopy is the gold standard for visualization and treatment of intrauterine pathology. Because hysteroscopic interpretation remains operator-dependent, artificial intelligence (AI) has been evaluated as a tool to improve consistency, lesion recognition, and decision support. We aimed to systematically review AI, machine learning (ML), deep learning (DL), or computer-aided diagnosis (CAD) applications in hysteroscopy. Methods: A systematic search of PubMed/MEDLINE and EBSCOhost was performed from database inception to 8 March 2026, supplemented by targeted searches. Risk of bias was assessed using QUADAS-2 (diagnostic), PROBAST (prognostic), RoB2, and structured technical quality domains. Results: Nineteen primary studies were included, covering five areas: diagnostic classification and object detection (n = 8), real-time lesion detection and localization (n = 4), segmentation and visual-field support (n = 3), operative guidance (n = 1), and prognostic or decision-support applications (n = 3). Performance was highest in narrowly defined binary tasks and in large multicenter systems (e.g., ECCADx: AUC 0.979 internal, 0.975 external) and in prognostic fertility-prediction models after hysteroscopic adhesiolysis (AUC up to 0.992). Broader multiclass classification of heterogeneous lesions showed uneven and lower performance. Most studies were single-center, retrospective, and lacked external validation. Only one randomized study linked AI support to measurable procedural outcomes. Conclusions: The available studies indicate good technical performance in selected hysteroscopic tasks, particularly binary classification, focal lesion detection, and postoperative fertility stratification.

Current evidence, however, remains limited by retrospective design, operator-dependent image acquisition, inconsistent validation, and scarce outcome-based clinical testing. In the short term, the most likely role of these systems is to support image interpretation, improve visual quality control, highlight suspicious lesions, and integrate hysteroscopic findings with complementary clinical data.

Keywords: hysteroscopy; hysteroscopic; gynecology; artificial intelligence; deep learning; machine learning; computer-aided diagnosis; computer vision; neural networks; convolutional neural network

Conclusion:
The existing literature shows that machine-learning and deep-learning systems can classify endometrial lesions, detect focal pathology in real time, segment relevant intraoperative structures, and support selected prognostic decisions. The strongest evidence currently concerns binary diagnostic classification for AEH and endometrial cancer, focal lesion detection, and fertility-related risk stratification after hysteroscopic treatment of intrauterine adhesions. At the same time, the field remains constrained by retrospective design, operator-dependent image generation, limited external validation, heterogeneous reference standards, and scarce evaluation of patient-centered outcomes. Unlike laparoscopy, where AI-robotics integration is moving toward real-time guidance and image overlay, hysteroscopy remains a manually performed procedure. Near-term use is therefore likely to remain concentrated in image processing, texture analysis, multimodal data integration, and decision-support. These systems are most likely to support perception by improving interpretation consistency, highlighting suspicious lesions, assisting biopsy targeting, and integrating hysteroscopic findings with clinical data. Translating these technical achievements into clinical practice requires prospective multicenter validation, standardized datasets, and workflow integration.

Discussion:
This review synthesized nineteen primary studies applying AI, ML, and DL to hysteroscopy across diagnostic, technical-support, operative, and prognostic tasks. Across these studies, performance is strongest in large binary diagnostic systems

for AEH and endometrial cancer and in narrow prognostic models for postoperative fertility after adhesiolysis. The ECCADx system, achieving an AUC of 0.979 on internal validation and 0.975 on external multicenter data, is among the highest reported in AI-based gynecological diagnostics [54]. At the same time, the overall evidence base remains preliminary. Most studies were retrospective, most were single-center, and only one directly evaluated a clinical procedural endpoint in a comparative design [50]. The available evidence therefore establishes technical feasibility in selected tasks but does not yet demonstrate prospective clinical effectiveness or workflow superiority in routine practice, neither of which can be inferred from retrospective, single-center accuracy estimates.

A consistent pattern across the reviewed literature is the difference between performance in narrow, well-delimited tasks and performance in broader multiclass classification. In Zhang et al. [44], binary discrimination between benign and premalignant/malignant lesions outperformed five-class classification within the same dataset. In Raimondo et al. [53], overall accuracy remained acceptable (87%), but recall for the preneoplastic/neoplastic category fell to 28%, meaning that more than 70% of cases in this category were not detected as such by the model.

The best-supported hysteroscopic applications are image-based recognition tasks: lesion classification, focal lesion localization, and detection of subtle inflammatory patterns. These are also tasks in which human interpretation varies substantially with experience, image quality, and clinical context [14,16,18,19]. At the current stage, the most plausible role of AI is not “replacement” of the hysteroscopist, but support for more consistent recognition of patterns that may otherwise be overlooked, overcalled, or interpreted differently across operators. Chronic endometritis (CE) is a good example of a setting in which AI may be clinically useful despite only partially overlapping hysteroscopic, histopathological, immunohistochemical, and microbiologic reference standards [65]. The CNN for automatic detection of endometrial micropolyps associated with CE developed by Kitaya et al. [51] achieved an AUC of 0.930, comparable to experienced gynecologists.

Examples of AI narrowing the performance gap between junior and more experienced examiners are also available from gynecological ultrasonography. In the study by Xu et al., a YOLOv8-based model for deep infiltrating endometriosis improved the AUROC of two junior sonologists from 0.748 to 0.878 and from

0.713 to 0.798, with sensitivity increasing up to 94.35% [66]. This supports the broader point that AI may be particularly useful in operator-dependent visual diagnostics, where expert-level pattern recognition is difficult to acquire and immediate supervision is not always available. Deep learning has also been applied to 3D transvaginal ultrasound for intrauterine adhesions, reaching high accuracy with external validation [67]. For T-shaped uterus, marked discordance between competing 3D-ultrasound diagnostic criteria illustrates the instability of morphometric thresholds [68], while a recent machine-learning model based on quantitative 3D-TVUS parameters achieved good diagnostic discrimination [69]. At the same time, poor-to-moderate inter-observer agreement for a proposed gynecological TVS image-quality scoring system shows that variability begins already at image acquisition and labeling [70].

Hysteroscopic image quality depends heavily on the procedure itself. Distance from tissue, viewing angle, bleeding, specular reflections from the fluid-distension medium, variable cavity distension, motion blur, and the curved, partially occluded field of view all shape the input available to the model. This differs from colonoscopy and gastroscopy, where larger benchmark datasets and better-developed artifact frameworks have supported more generalizable systems [71,72,73,74]. In hysteroscopy, the studies by Wang et al. on bubble segmentation [47], by Burai et al. [43] on uterine-wall extraction, and the myomectomy plane segmentation by Török and Harangi [42] address technical conditions that are directly relevant for practical deployment.

Translational adoption also depends on clinically interpretable output and reliable uncertainty estimates at the point of care. The deep-learning models that perform best on hysteroscopic images are largely opaque, and the resulting “black-box” character, together with the need for interpretability, calibrated confidence, and demonstrable fairness, is now recognized as a principal barrier to clinical deployment of AI in medical imaging [24]. None of the included hysteroscopic studies reported calibration of predicted probabilities or provided interpretability output that a clinician could use to accept or override a given prediction. For real-time hysteroscopy, retrospective expert-level accuracy is insufficient unless the output is interpretable and its uncertainty is clearly communicated during the examination. Confidence calibration and interpretable localization are needed before these systems can support intraoperative decisions.

Hysteroscopy, at its current developmental stage, remains a manually performed procedure. Navigation, lesion exposure, tissue handling, instrument positioning, and the final operative decision still depend on the operator's real-time judgment. This limits direct comparison with laparoscopy, especially robotic laparoscopy, where the operative interface is already digital and telemanipulated, preoperative imaging can be overlaid onto the operative field, critical structures can be annotated in real time, and workflow analysis can be integrated more directly into surgical guidance [27,75,76,77]. Hysteroscopy does not currently offer an equivalent platform. The main contribution of AI is more likely to lie in perception support, image-quality control, lesion highlighting, multimodal integration, and postoperative prediction.

The reviewed studies point to four near-term applications: (1) image processing and classification during or after the procedure, as a decision-support layer that augments interpretation without replacing manual control [53,54,55]; (2) texture and vascular-pattern analysis for subtle pathology, including chronic endometritis [51], early hyperplasia [44,54], and structural anomalies; (3) multimodal integration with 3D ultrasound or sonohysterography for pre-hysteroscopic planning in T-shaped uterus, uterine septa, and Asherman syndrome [52,59], where cavity-level and panoramic imaging provide complementary information; and (4) AI-assisted postoperative outcome prediction, as demonstrated by the Beijing group [52,56,57,59], in which hysteroscopic findings contribute to probabilistic clinical decision-support tools.

Limitations:

Risk-of-bias assessments consistently show that technically sound model development is undermined by limited validation. This pattern is not specific to hysteroscopy, as several clinical studies evaluating AI in surgery rely on sparse multicenter data, heterogeneous annotation standards, limited external validation, and insufficient outcome-oriented evaluation [78]. Fourteen of the nineteen included studies were retrospective and single-center. Many used internal train–test splits from the same institutional archive. Under these conditions, performance may be inflated by shared acquisition settings, equipment, annotation habits, or patient mix [78,79,80,81,82]. This problem is compounded when the unit of analysis is the image or frame rather than the patient. If multiple images from the same patient are distributed across training and test sets, effective sample size is overstated and leakage may remain undetected.

A further constraint concerns the provenance of the datasets themselves. Most included studies drew on curated retrospective archives in which low-quality frames were excluded before model development, so that the images used for training and testing represent favorable acquisition conditions instead of routine hysteroscopic environment, in which glare, bleeding, mucus, debris, motion blur, and incomplete cavity distension are common. Annotation reference standards were heterogeneous and frequently rested on a single expert reader or on the interpreting clinician, without reported inter-annotator agreement, so that label noise cannot be quantified. This combination of curated inputs and single-source labels limits ecological validity, as reported performance reflects development conditions rather than real-world deployment.

The heterogeneity of reporting across studies makes cross-study comparison unreliable. Studies report different primary metrics (AUC, accuracy, F1, Dice, C-statistic), use different reference standards (histopathology from directed biopsy, expert hysteroscopic annotation, blinded panel review), and employ different units of analysis (per-image, per-frame, per-patient). The choice of per-image instead of per-patient analysis artificially inflates effective sample sizes and can mask patient-level confounding: when multiple images from the same patient are split across training and test sets—a form of data leakage whose performance-inflating effects have been quantified at 30–55% in CNN-based classification tasks [81,82]—reported performance metrics are optimistic in ways that are practically impossible to detect without access to the original data. For the detection and localization studies specifically, the unit of analysis, the intersection-over-union threshold defining a true detection, and whether negative frames or videos were included in the test set were reported inconsistently; this incomplete reporting limits comparison across studies. Inter-observer variation in histopathological diagnosis of endometrial hyperplasia remains substantial even among specialist pathologists [83], compounding the difficulty of constructing unambiguous reference standards for AI training. The study by Raimondo et al. [53] illustrates this limitation: overall classification accuracy of 86.74% coexisted with a recall of only 27.59% for the preneoplastic/neoplastic class, precisely the category of greatest clinical relevance and the one for which a missed diagnosis carries the highest patient risk.

The same limitation appears differently in the prognostic studies. Their reported AUCs are high, but validation remains internal to a single cohort line [52,56]. Calibration of predicted probabilities was rarely reported. Li et al. [56] included decision-curve analysis and reported a net benefit of 69.4% for the fertility-assessment system, whereas Li et al. [52] reported risk-stratified ART benefit without formal probability calibration. Earlier Cyprus CAD work provided an internally validated texture-based proof of concept [41]. The Beijing intrauterine-adhesion models report high AUCs, but validation remains internal to one cohort line [52,56]. Wang et al. [54] provide the strongest diagnostic validation with independent multicenter test datasets; Chen et al. [50] remains the only study with a comparative procedural endpoint. The imbalance between tasks where AI implementation produces statistically significant numbers and their clinical utility can be illustrated by two studies addressing hysteroscopic myomectomy [50,57]. The work of Chen et al. [50] remains the only randomized comparison linking AI support to a procedural endpoint. It reports a shorter operative time (around 10 min) and statistically lower blood loss despite identical median values in both groups and only slightly different interquartile ranges. The clinical relevance, however, remains speculative. First, the almost identical and extremely low blood loss contrasted with a difference of approximately 25% in surgical time. It is surprising, because surgical time is a strong predictor of blood loss at least in, e.g., laparoscopic myomectomy [84,85]. By contrast, Givon et al. [57] addressed the clinically relevant problem of incomplete myomectomy. In real-world settings, incomplete myomectomy has been reported in 36.69% of hysteroscopic approaches and is associated with complications (PR: 2.77; 95% CI: 1.43–5.38) [86]. The model performance in [57] was clearly lower than in the Beijing fertility studies or ECCADx, but the results suggest that AI can help identify cases in which complete hysteroscopic treatment is less likely.

Search Set 7

Key word: IVF success prediction clinical data

Paper No:1

Title:
The prospect of artificial intelligence to personalize assisted reproductive technology

Year: Published: 01 March 2024

Abstract:
Infertility affects 1-in-6 couples, with repeated intensive cycles of assisted reproductive technology (ART) required by many to achieve a desired live birth. In ART, typically, clinicians and laboratory staff consider patient characteristics, previous treatment responses, and ongoing monitoring to determine treatment decisions. However, the reproducibility, weighting, and interpretation of these characteristics are contentious, and highly operator-dependent, resulting in considerable reliance on clinical experience. Artificial intelligence (AI) is ideally suited to handle, process, and analyze large, dynamic, temporal datasets with multiple intermediary outcomes that are generated during an ART cycle. Here, we review how AI has demonstrated potential for optimization and personalization of key steps in a reproducible manner, including: drug selection and dosing, cycle monitoring, induction of oocyte maturation, and selection of the most competent gametes and embryos, to improve the overall efficacy and safety of ART.

Conclusion:
With respect to ART, several groups have developed CDSS frameworks or decision-making tools for use at key decision-points in the clinic, and/or embryology laboratory17,30,31. Personalization in further avenues could better improve the clinical outcomes of ART. Ovarian response has been shown to vary significantly depending on ovarian reserve, between ethnic groups114,115, FSH receptor genetic polymorphisms116, and body weight19,117. Therefore,

incorporating such factors which influence pharmacokinetic parameters when dosing gonadotropins9,19,20, or suppressing premature ovulation20,118, may be beneficial. ML methods could also help tailor luteal phase support regimens to certain patient subgroups, where a lack of clinical consensus currently exists119.

The ubiquity of electronic health records (EHRs) has accelerated the development of CDSSs15. A predominant barrier to adoption is trustworthiness, especially with 'black-box' AI systems29, which has led to transparency being a key characteristic preferred by clinicians as such models offer simpler interpretations, although may compromise accuracy when applied to more complicated learning tasks28. Implementations of 'black-box' models are evolving, especially for embryological analyses, due to the data being primarily image-based; in turn, efforts in explainability have emerged to seek insights for model generalizability, fairness, and trustworthiness94,95. Misleading conclusions may be reached if clinical inference is neglected during the decision-making process since such methods are often correlation-based and prone to 'overfitting'120. Generating counterfactual examples in this context, such as: "what if the optimal TD was yesterday(?)", or "what if the other embryo were implanted(?)", are generally unavailable—and to further exacerbate this—ground truths are often based upon clinical guidelines/scoring rather than objective outcome labels. The emergence of omics analyses offers an alternative, and arguably more efficient, solution for clinical and embryological assessment, although advancements currently remain of a preliminary nature18,108. Ultimately, appropriate assessment of CDSSs for ART is necessary in practical, ethical, and clinical contexts prior to clinical adoption. Rigorous validation with comprehensive standardized reporting is essential for establishing trustworthy models before attempting viable integration into clinical workflows21,121. Research conduct and reporting guidelines such as PROBAST-AI are in progress for the wider field of AI for healthcare, and with this at hand, a more granular and contextual guideline for AI in the domain of ART can be proposed122,123.

Salient efforts from both academia and industry have validated the utility of retrospective data to enable data-driven decision-making for ART123. To ensure viable deployment, these models can benefit from larger, multi-center datasets that incorporate both heterogeneous patient populations and also capture the idiosyncratic nature of clinical practice worldwide. Achieving this is best achieved through a collaborative effort from all stakeholders representing multiple disciplines across the AI and healthcare landscape21. Furthermore, streamlining workloads is an essential objective of CDSSs, and seamless implementation with, or within, EHR systems are essential to not inadvertently decrease the efficiency of clinical workflows. Prospective validation (e.g., well-designed RCTs) with relevant outcome measures is a key step to assess the efficacy and efficiency of these models in clinical environments and thus demonstrate impact on patient outcomes. With such efforts in place, a comprehensive end-to-end CDSS seems a plausible future goal. Whether this paradigm should extend to an autonomous AI clinician within the ART domain remains an open and contentious question. The use of AI to automate some of the tasks currently performed by clinicians or laboratory staff could have implications in training and a potential loss of expertise in the workforce, but may also free up staff time to focus on more challenging and physically demanding technical processes. Reflections on the current literature to date elicit valuable questions regarding future studies, including determining the specification of what should be measured/captured, to what precision, and how often. Decision points cannot necessarily be considered in isolation, and the relationships between some of the key topics described in this review require further interdisciplinary research to prioritize the individualization and utility of certain decisions over others. The intersection of AI and ART undoubtedly remains a nascent and valuable field of study, which has the potential to reduce intensive resources, whilst ultimately improving clinical outcomes for patients

Paper No:2

Title:
Artificial intelligence and assisted reproductive technology: A comprehensive systematic review

Year: January 2025

Abstract:
The objective of this review is to evaluate the contributions of Artificial Intelligence (AI) to Assisted Reproductive Technologies (ART), focusing on its role in enhancing the processes and outcomes of fertility treatments. This study analyzed 48 relevant articles to assess the impact of AI on various aspects of ART, including treatment efficacy, process optimization, and outcome prediction. The effectiveness of different machine learning paradigms—supervised, unsupervised, and reinforcement learning—in improving ART-related procedures was particularly examined. The findings indicate that AI technologies significantly enhance ART processes by refining tasks such as embryo and sperm analysis and facilitating personalized treatment plans based on predictive modeling. Notable improvements were observed in the accuracy of diagnosing and predicting successful outcomes in fertility treatments. AI-driven models provided more precise forecasts of the optimal timing for clinical interventions such as egg retrieval and embryo transfer, which are critical to the success of ART cycles. The integration of AI into ART represents a transformative advancement, substantially improving the precision and efficiency of fertility treatments. The continuous evolution of AI methodologies is likely to further revolutionize this field, enabling more tailored and successful treatment approaches. AI is becoming an indispensable tool in reproductive medicine, enhancing both the effectiveness of treatments and the clinical decision-making process. This review underscores the potential of AI to act as a catalyst for innovative solutions in the optimization of ART.

Conclusion:

As AI makes its indelible mark on ART, it catalyzes an era where the synergy between cutting-edge technology and medical expertise is not just beneficial but essential. AI's role in ART extends to optimizing protocols across various stages of IVF, from ultrasound imaging for follicle assessment to precise gonadotropin dosing and identifying ideal sites for embryo transfers. In the IVF lab, AI excels in micromanipulation techniques, enhancing the precision of procedures such as ICSI and assisted hatching, and it aids clinicians in selecting the optimal embryos for transfer or cryopreservation.

Adopting AI requires a balanced approach, recognizing it as an ally rather than a competitor. It's a partnership where AI's efficiency complements clinicians' expertise, making the promise of higher ART success rates attainable. This future-oriented vision of ART underscores a collaborative approach to AI, advocating for mastery over fear of displacement. By embracing AI, fertility care professionals can redefine patient outcomes, ensuring that the field remains at the forefront of innovation and care excellence. This integration of AI into ART is a testament to a collaborative commitment to advancing reproductive health, grounded in the ethos of enhancing human capability with machine precision.